



Department of Education

STEM Education
National Schools of Excellence
Biology Teachers Resource Book
Grades 11
Term 2

DOMAIN PHYLA

Science **Biology**

Technology

Engineering

Mathematics

Issued free to schools by the Department of Education

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ISBN

Acknowledgements

This STEM Biology Teacher Resource Book for Grade 11 constitutes ten (10) Thematic Areas (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented.

Ultima Thulë Ltd was engaged by the Department of Education to develop the STEM Curriculum. The content was generated by writers from universities, secondary schools, new university graduates with knowledge in the subject matter and industry subject experts. Quality assurance groups and the Science Subject Advisory Committee have also contributed to the development of this syllabus.

This document was developed with support from the General Education Services Division of the Department of Education.

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SECRETARY'S MESSAGE

This Biology Teacher Resource Book is to be used exclusively by teachers of Biology to teach Grade 11 Students enrolling for the STEM Education through the Schools of Excellence throughout Papua New Guinea. The STEM Biology Bridging Syllabus ensured that Grade 11 students enrolling for STEM Education were grounded in the basic concepts espoused in Biology before undertaking the advanced concepts introduced in this Teacher Resource Book.

The Teacher Resource Book conforms to the visions of the National Education Plan and the School of Excellence Policy. Furthermore, it is undergirded by a policy and legislative matrix that were issued by the National Government over the last decades, such as the Vision 2050 and Development Strategic Plan 2030. The Preamble of the National Constitution, especially the First Directive of the *National Goals and Directive Principles* makes a clarion call for the Integral Human Development, which the Department took heed of to ensure, through this new syllabus, that *all forms of beneficial creativity, including sciences and cultures, [are] to be actively encouraged*. A principle object of the *National Education Act 1983 (amended 1995)* reiterates this virtue of Integral Human Development (Section 4).

In accordance with the widespread demand for a home-grown curriculum to replace the existing one with the view to position the country as a bastion of innovation in the new century the Secretary of Education has embarked on introducing the STEM Curriculum to the Schools of Excellence. It is hoped that students coming through the STEM Education will not only gain productive employment in the knowledge-based economy but become creators, innovators and inventors of knowledge and ultimately commercialize their creativity through development of their intellectual property rights.

In addition, there are four core elements espoused in this syllabus, insofar as student development is concerned: Students are expected to gain knowledge and skills and apply them to solve every day problems whilst appreciating Biology as a fundamental field of science.

This STEM Biology Teacher Resource Book for Grade 11 constitutes ten (10) Thematic Area (Thema) and is intended for students enrolling for STEM Education. In this booklet only three Thema are presented. By studying these Thema students are expected to gain the foundational knowledge needed in Biology.

The success of this new Curriculum is highly dependent on the Teachers. They play a pivotal role by being innovative and keeping abreast of new information on scientific research and innovations. The challenge for teachers of Biology is to engage students to learn realistic context for increased and better understanding and appreciate the relevance of Biology to human lives, including good health and environment.

In order to ensure Teachers discharge their responsibilities adequately this Teacher Resource Book is accompanied by a Teacher's Guide and Teacher's Resource Book which provides sufficient teaching and assessment materials to help Teachers.

I commend and approve this Teacher Resource Book as the official curriculum for Biology to be used for the Grade 11 STEM Education in the Schools of Excellence throughout Papua New Guinea.

DR UKE KOMBRA, PHD
Secretary for Education

INTRODUCTION

Biology is the study of living organisms and how they interact at multiple levels to sustain life. The basic building block of life is considered to be the cell. The hereditary units are the genetic sequences and these are encased within the cells. This genetic information ensures the hereditary traits necessary for survival of the species is passed on from parents to progenies. Biology is divided into different fields of study. For instance, botany is the study of plants whilst zoology involves the study of animals. Micro-organisms are studied in microbiology. Recent fields that emerged in the last few decades include molecular biology, genomics, and so on. Since Biology is an experimental science, laboratory work becomes an essential part of the syllabus. Practical work can be used to enable students to investigate the properties and reactions of substances and to provide opportunities for students to learn and test the principles and concepts of Biology. Since Biology is a highly specialized subject students undertaking Biology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills. The modules in this Teacher Resource Book are sequenced to guide teachers of Biology through the process of imparting basic Biology knowledge and skills to students.

The Grade 11 Biology Teacher Resource Book entails ten (10) Thematic Areas (Thema) and these are listed below. Each Thema has a single Module and three topics. Over time different Modules will be introduced.

Since Biology is a highly specialized subject students undertaking Biology are expected to possess high levels of cognitive and numeracy competency, and a firm grasp of the language skills.

The modules in this Teacher Resource Book are sequenced to guide teachers of Biology through the process of imparting basic Biology knowledge and skills to students.

THEMA	MODULE	TOPICS
BIODIVERSITY	Agricultural Biodiversity	1.Agricultural Biodiversity in Papua New Guinea 2.Genetic Diversity
	Ornithology	1. Birds of Paradise 2. Inspirations from bird behaviour
BIOTECHNOLOGY	Concepts and Applications of Biotechnology	1.Recombinant DNA Technology 2.Industrial Biotechnology
HEALTH AND MEDICINE	Medicine and Health Sciences	1.Pharmacology 2.Immunization and Vaccines 3.Public Health
EARTH SCIENCE	Environment	
WASTE MANAGEMENT	Biological Wastes	
INFRASTRUCTURE	Natural Inspirations in Infrastructure	
	Agricultural Infrastructures	

ENERGY	Cell Energy and Power	1. Photosynthesis 2. Catabolism
BATTERY TECHNOLOGY	Battery Technology Application in Biological Systems	
INFORMATION SYSTEM		
COMPUTER SCIENCE		
ARTIFICIAL INTELLIGENCE	Biological Concepts in Artificial Intelligence	1. Neurobiology 2. Neurobiological models in Artificial Intelligence
ENTERPRENEURSHIP		1. Intellectual Property (Patents, Copyrights, Trademarks, etc) 2. How to protect Intellectual Property 3. How to commercialize Intellectual Property

RATIONALE

There is a confluence of events that necessitated the need to introduce STEM Curriculum to the Schools of Excellence:

1. The School of Excellence Policy that was launched by Prime Minister Honourable James Marape, MP, in June 2020, calls for the establishment of the Schools of Excellence and the introduction of STEM Curriculum;
2. The Government has decided to replace the Observation Based Education to Standards Based Education in 2019 and tasked the Department of Education to introduce the STEM Curriculum as part of its reform agenda;
3. There is public outcry, both from public and private sectors, over the quality of graduates coming through the National Education System; and
4. The changing landscape in the technology domain globally and the pressing need to produce the next generation of workforce that is agile and can adopt to the changing circumstances as well as become enterprising individuals through their creation of intellectual properties.

The combination of these factors has led the Department of Education to adopt the STEM Curriculum and to introduce this to the Schools of Excellence as a necessary disruptive intervention in an effort to develop the next generation of national intellectual capital and to drive the growth and prosperity of the country through creation of intellectual properties.

In developing the STEM Curriculum, it was realized that the Grades 9 and 10 Science Syllabi does not provide a firm foundation to launch students into STEM Curriculum at the Schools of Excellence. Consequently, a STEM Bridging Program was necessary.

In the Biology Teacher Resource Book students were introduced to the basic concepts and knowledge of Biology in order for them to develop an appreciation of the discipline in a broader context. In this Teacher Resource Book students are introduced to advanced concepts in Biology as they are applied to the ten Thema.

The introduction of the STEM Curriculum, especially the Bridging Program, will cause consequential changes to the Upper Classes of STEM Education and the Lower Secondary Syllabi. This Teacher Resource Book will be introduced in the course of time.

NEW NOMENCLATURE

As part of the overall Curriculum Development new nomenclature were introduced. These are explained below:

Term	Explanation
Domain	Each STEM area is defined as <i>domain</i> (e.g. Science Domain, Technology Domain, etc.)
Phylum	What would normally be regarded as “subject” (e.g. Biology, Physics, Chemistry, etc.) is now regarded as <i>phylum</i> ; the plural is <i>phyla</i> .
Thema	Thematic areas of studies
Module	A specific concept within the Thema which is being studied
Topic	A specific aspect of the Module that is being studied

STEM Domains

At this juncture of our national development there is no other task before us more important than deploying a robust STEM Curriculum that develops the next generation of highly educated students to enter the work force. By building on the best of current practices and standards the STEM Curriculum aims to take us, beyond the constraints of the existing structures of schooling, towards a shared vision of excellence nationally and marketing the nation globally in terms of Science, Technology, Engineering and Mathematics. In this STEM Curriculum, there are ten (10) Thema (Thematic Areas) introduced.

Science

It is the study of the natural world, including the laws of the nature associated with physics, chemistry and biology and the treatment or application of facts, principles, concepts or conventions associated with these disciplines. Science is both a body of knowledge that has been accumulated over time and a process-scientific inquiry-that generates new knowledge. Knowing from science informs the engineering design process. In this STEM Curriculum the Science Domain has three Phyla – Biology, Chemistry and Physics.

Technology

Comprises the entire system of people and organizations, knowledge, processes, and devices that go into creating and operating technological artifacts, as well as the artifacts themselves. Throughout history humans have created technology to satisfy their wants and needs. Much of modern technology is the product of science and engineering and technological tools used in both fields. There are two Phyla – Information Technology and Computer Sciences – introduced in this STEM Curriculum for the Technology Domain.

Engineering

Concerns both the knowledge about the design and creation of products and a process for solving problems. The design process must result in a final product that must operate or function under various constraints. One constraint in engineering design is the laws of nature. Engineers must learn to harness the natural forces at work around them and through their design process develop products that must operate in harmony with nature. Other constraints include things such as time, money, available materials, ergonomics, environmental regulations and manufacturability. Engineering utilizes concepts in science and mathematics as well as technological tools. There are three (3) Phyla introduced in the Engineering Domain – Civil, Mechanical and Electrical.

Mathematics

It is the study of patterns and relationships among quantities, numbers, and shapes. Specific branches of mathematics include arithmetic, geometry, algebra, trigonometry and calculus. Mathematics is used in science and in engineering. In this STEM Curriculum, Mathematics is considered as a Service Domain; specific topics tailored towards specific Thema are also considered relevant to undergird the relevance and applicability of Mathematics Domain to other fields of knowledge.

STEM BIOLOGY CONTENT OVERVIEW – TERM 2

Table 1.1: Thema Sequence and Duration

Grade 11 Term 2: STEM-proper Course		
BIODIVERSITY (3 Weeks)	1. Agricultural Biodiversity	1. Agricultural Diversity in Papua New Guinea 2. Genetic Diversity
	2. Ornithology	1. Bird of Paradise 2. Inspirations from Bird Behaviour
BIOTECHNOLOGY (3 Weeks)	1. Concepts and Applications of Biotechnology	1. Recombination DNA Technology 2. Industrial Biotechnology
HEALTH AND MEDICINE (3 Weeks)	1. Health and Medical Biology	1. Pharmacology 2. Immunization and Vaccines 3. Public Health

Table 1.2 Sequencing & Benchmarking STEM Chemistry Content Standards

Setting STEM Chemistry Content Standards for Progressive Learning & Assessment			
Thema	Biodiversity	Broad Content Standard	
Module 1	Agricultural Biodiversity	11.1: Recognize and appreciate the rich biodiversity within Papua New Guinea and how to industrialize our biodiversity cautiously and conservatively from outside influence.	
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks
1.Agricultural Diversity in Papua New Guinea	11.1.1a: Describe the four levels of organisation of biodiversity. 11.1.1b: Describe the diverse agricultural biodiversity of Papua New Guinea 11.1.1c: Identify and describe the wild relatives of agricultural crops native to Papua New Guinea 11.1.1d: Identify and describe the limitations to agricultural development in Papua New Guinea.	Students master the ability and knowledge to: - Identify and describe the arms of agricultural biodiversity that contribute genetic resources for food and agriculture as well as ecosystem support services. - Evaluate and analyze the vast genetic diversity within Papua New Guinea and by what means to conserve it as well as to manipulate it utilizing technology for	Research Project + Written Test

2. Genetic Diversity	11.1.2a: Identify and describe the genetic diversity of native agricultural crops in Papua New Guinea compared to the world and how to conserve native species. 11.1.2b: Recognize the importance of genetic sequencing of native agricultural crops in PNG and their applications to address climate resilience and food security.		purposes of food security and enhancing productivity.	
Thema	Biodiversity	Broad Content Standard		
Module 2	Ornithology	11.2: Understand the significance of Birds of Paradise to Papua New Guineans and glean the importance of the behaviours of birds and their implications in geopolitics and organizational strategies		
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks	
1. Birds of Paradise	11.2.1a: Analyze the importance of the Ornithology of the Birds of Paradise as our National Emblem, the species diversity and their natural character or behaviour.	Students' ability to demonstrate knowledge and skills in; - Describe the ornithology of the Birds of Paradise species and their significance in Papua New Guinea. - Discuss and analyze the geopolitical significance of inspirations from bird behaviors that are unique to human or organizational strategies	Research Project + Written Test	
2. Inspirations from Bird Behaviour	11.2.2a: Analyze the nesting and predatory behaviour of birds that inspire geopolitical strategies utilized by organizations. 11.2.2b: Appreciate the inspirations from bird behaviour that has significant applications in diverse settings. 11.2.2c: Give examples: Mendondi and Tenduage Strategy as applied in Organizational and Corporate Strategies.			
Thema	Biotechnology	Broad Content Standard		
Module 1	Concepts and Applications of Biotechnology	11.1: Appreciate the concepts and applications of biotechnology in the medical or pharmaceutical and agricultural industries.		
Topics	Performance Indicators	Module Benchmark	Assessment Tasks	

	Students are expected to:		
1. Recombination of DNA Technology	11.1.1a: Define and discuss the principle concepts of recombinant DNA technology. 11.1.1b: Describe the basic tools and techniques utilized in recombinant DNA technology.	Student's master the ability and knowledge to: - Analyze and interpret the basic concepts of biotechnology - Generally describe the concept of recombinant DNA technology and applications - Evaluate and appreciate how industries are utilizing biotechnologies to improve production.	Written Test
2. Industrial Biotechnology	11.1.2a: Identify and describe the applications of recombinant DNA technology in the medical or pharmaceutical and agricultural industries. 11.1.2b: Appreciate the contribution of biotechnology in the improvements and advancement of health care and crop production.		

Thema	Health and Medicine	Broad Content Standard	
Module 1	Health and Medical Biology	11.1: Identify and generally describe the key areas in medical and health sciences.	
Topics	Performance Indicators Students are expected to:	Module Benchmark	Assessment Tasks
1. Pharmacology	11.1.1a: Describe how a drug is being administered, absorbed, metabolized and excreted from the body. 11.1.1b: Interpret how a drug works and its side effects	Students master the ability and knowledge to: - Describe the basic concepts of pharmacology - Describe the different vaccines and - Describe the SARS-CoV2-2019 Pandemic from the public health perspective and its impacts (health, economic, social and psychological) at a global and local context.	Written Test
2. Immunization and Vaccines	11.1.2: Describe the different vaccines and their mechanism of action.		
3. Public Health	11.1.3: Interpret and describe the public health aspects and impacts (Health, economic, social and psychological) of SARS-CoV2–2019 (COVID-19) Pandemic.		

Table 1.3 Mapping Content Progressions by Modules

Thema/Module	Broad Content Standard	11.1	11.2	11.1	11.1
Biodiversity/ 1. Agricultural Biodiversity	11.1: Recognize and appreciate the rich biodiversity within Papua New Guinea and how to industrialize our biodiversity cautiously and conservatively from outside influence.	✓			

Biodiversity/ 2. Ornithology	11.2: Understand the significance of Birds of Paradise to Papua New Guineans and glean the importance of the behaviours of birds and their implications in geopolitics and organizational strategies		✓		
Biotechnology/ 1. Concepts and Applications of Biotechnology	11.1: Appreciate the concepts and applications of biotechnology in the medical or pharmaceutical and agricultural industries.			✓	
Health and Medicine/ 1. Health and Medical Biology	11.1: Identify and generally describe the key areas in medical and health sciences.				✓

THEMA BIODIVERSITY

MODULE 11.1: AGRICULTURAL BIODIVERSITY

Topic 1: Agricultural Biodiversity in Papua New Guinea

Key Concepts that a Student should acquire:

1. *Understanding of biodiversity*
2. *Understand the different levels of biodiversity*
3. *Appreciate Papua New Guinea's biodiversity*
4. *Appreciate Papua New Guinea as one of the world's oldest agriculture civilization*
5. *Appreciate Papua New Guinea's agricultural biodiversity (native crops)*
6. *Know the limit to Papua New Guinea's agricultural development*
7. *Describe the wild relatives of agricultural crops*
8. *Understand the genetic diversity of Papua New Guinea's native crops*
9. *Understand the factors influencing the genetic diversity of native crops*
10. *Be able to describe the genetic diversity of other native crops*
11. *Recognize the importance of genetic diversity and how it can be manipulated for economic development*
12. *Understand that ornithology is the study of birds*
13. *Appreciate the diversity of the Birds of Paradise and their significance to Papua New Guinea*
14. *Gain some inspirations by studying bird behavior and how they can be applied in geopolitics and corporate settings*

Activity 1: Classroom Activity

Title: General Biodiversity

(1) Define and describe generally what biodiversity is?

What is biodiversity? - Biodiversity is short for biological diversity. It includes all organisms, species, and populations; the genetic variation among these; and all their complex assemblages of communities and ecosystems. It also refers to the interrelatedness of **genes**, **species**, **ecosystems** and **functions** and their interactions with the environment. Biodiversity is a key measure of the health of any ecosystem, and of our entire planet. Every organism in an ecosystem, or biome, relies on other organisms and the physical environment. For example, plant and animal species need each other for food, and depend on the environment for water and shelter. Biodiversity describes how much variety an ecosystem has, in terms of resources and species and also genetically within species. A more diverse ecosystem will have more resources to help it recover from famine, drought, disease or even the extinction of a species. There are several levels of biodiversity, each indicating how diverse the genes, species and resources are in a region that enable functioning in an ecosystem.

(2) Define and describe the four levels of biodiversity?

Genetic Diversity

Genetic diversity is all the different genes contained in all individual plants, animals, fungi, and microorganisms. It describes how closely related the members of one species are in a given ecosystem. In simple terms, if all members have many similar genes, the species has low genetic diversity. Because of their small populations, endangered species may have low genetic diversity due to inbreeding. This can pose a threat to a population if it leads to inheritance of undesirable traits or makes the species more susceptible to disease. Having high genetic diversity helps species adapt to changing environments.

Species Diversity

Species diversity is all the differences within and between populations of species, as well as between different species. Every ecosystem contains a unique collection of species, all interacting with each other. Some ecosystems may have many more species than another. In some ecosystems, one species has grown so large that it dominates the natural community. When comparing the biodiversity of ecosystems, an ecosystem that has a large number of species, but no species greatly outnumbering the rest, would be considered to have the most species diversity. A large number of species can help an ecosystem recover from ecological threats, even if some species go extinct.

Ecosystem Diversity

Ecosystem diversity is all the different habitats, biological communities, and ecological processes, as well as variation within individual ecosystems. A region may have several ecosystems, or it may have one. Wide expanses of oceans or deserts would be examples of regions with low ecological diversity. A mountain area that has lakes, forests and grasslands would have higher biodiversity, in this sense. A region with several ecosystems may be able to provide more resources to help native species survive, especially when one ecosystem is threatened by drought or disease.

Functional Diversity

The way species behave, obtain food and use the natural resources of an ecosystem is known as *functional diversity*. In general a species-rich ecosystem is presumed to have high functional diversity, because there are many species with many different behaviors. Understanding an ecosystem's functional diversity can be useful to ecologists trying to conserve or restore damaged it, because knowing the behaviors and roles of species can point to gaps in a food cycle or ecological niches that are lacking species.

Functional Diversity is the biological and chemical processes such as energy flow and matter recycling needed for the survival of the species, communities and ecosystem.	Ecosystem Diversity is the variety of terrestrial and aquatic ecosystems found in an area on the earth.
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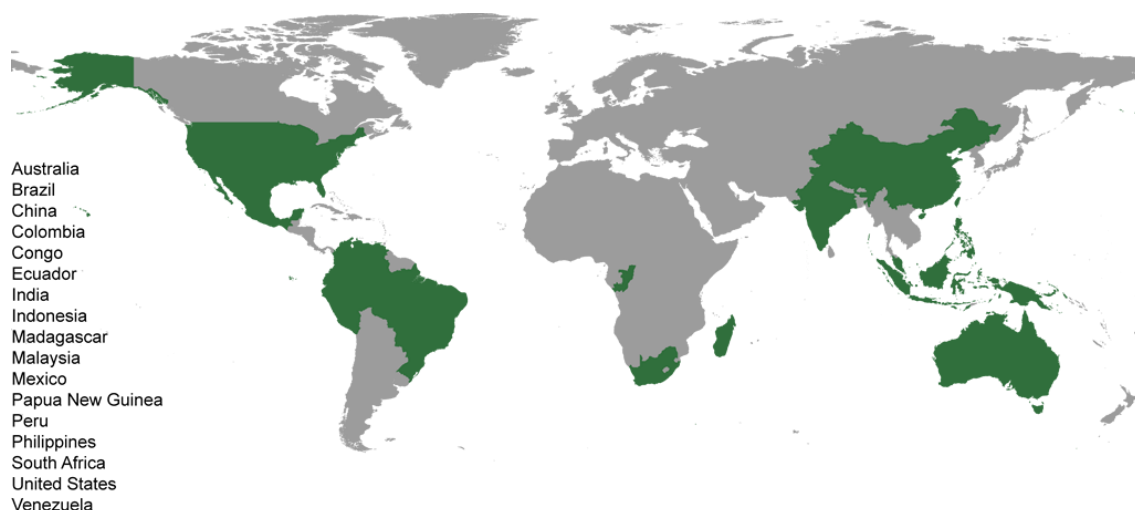
Genetic Diversity is the variety of genetic material within a species or a population.	Species Diversity is the number and abundance of species present in different communities.
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Figure BD-AB1: Representation of the four levels of biodiversity.
 (source: <https://www.slideshare.net/Anilkumar2120/concept-and-characteristics-of-biodiversity>)

(3) Describe how diverse the biodiversity of Papua New Guinea is?

Papua New Guinea's Biodiversity

Papua New Guinea occupies 1% of the global land area but hosts 5% of the world's total biodiversity. Two-thirds of the biodiversity (both flora and fauna) are endemic. PNG is considered as one of the world's megabiodiversity countries. The country is remarkably diverse in terms of species, landscapes and ecosystems.



Source: www.environment.gov.au/biodiversity/hotspots/index.html

Figure BD-AB2: Countries with megabiodiversity

The flora of Papua New Guinea is poorly known. Scientists estimate that more than half the plants and animals found in Papua New Guinea have yet to be scientifically named. Estimates for the number of vascular plant species for the entire island of New Guinea range from 11,000 to 25,000 species. The forests of the island of New Guinea constitute the third largest expanse of tropical rainforest on the planet, after the forests of the Amazon and the Congo Basin. Rainforests cover 28.2 million hectares of Papua New Guinea and comprise 80% of the forest estate, with the rest of the forest estate comprised of dry evergreen forest, swamp forest and mangroves. The total forest estate covers approximately 71% of the land area.

The richness of biodiversity in PNG is reflected in the following numbers:

- The number of plant species is estimated to be in the range of 15,000 to 20,000, which represent about 6% of world's flora.
- For animals, it is estimated that PNG has –
 - 150,000 species of insects;
 - 314 species of freshwater fishes (82 endemic);
 - 2800 species of marine fishes;
 - 641 species of amphibians and reptiles (328 endemic);
 - 740 species of birds (77 endemic); and
 - 276 species of mammals (69 endemic).

Extensive biodiversity is also found in the marine and aquatic environment in PNG. The country is part of the coral triangle, the area of the world's highest known marine biological diversity. Its coral reefs and associated marine habitat are home to about 2800 species of fishes, about 10%

of the world total. Almost all reef types found in PNG waters are within fringing and/or barrier reefs, with an estimated area of 40,000 km². In addition, the country has some of the largest unpolluted tropical freshwater systems in the Asia-Pacific region. Mangrove swamps occupy 51.6 million hectares (about 2% of the forest estate). There are 33 species of mangrove trees known, representing the highest mangrove diversity in the world.

It is estimated that there are about 1,035 different plant species being utilized for various purposes in PNG. Wildlife plays an important part in traditional diets, supplying the primary intake of proteins and fats in many highland areas and other isolated areas of the country. In coastal areas a wide variety of seafood, including fish, mollusks, and turtles, dominate local diets.

Biodiversity plays an important role in food and agriculture as the genetic resource base for seeds and planting materials of crops and breeds of livestock and fish for different types of crops, livestock and fish which the farmers can use for cultivation and farming to meet their food security, nutritional and socio-economic needs.¹

The rich biodiversity of the country is now being threatened by agro-forestry (oil palm plantations), logging, mining and other extractive industrial applications. The need for conservation has been prominent as of late.

¹ <http://www.fao.org/3/CA3422EN/ca3422en.pdf>

Activity 2: Classroom Activity

Title: Agricultural Biodiversity

(1) Define what agricultural biodiversity is?

*“Agricultural biodiversity is a broad term that includes all components of biological diversity of relevance to food and agriculture, and all components of biological diversity that constitute the agricultural ecosystems, also named agro-ecosystems: the variety and variability of animals, plants and micro-organisms, at the **genetic, species, ecosystem** and **functional** levels, which are necessary to sustain key functions of the agro-ecosystem, its structure and processes.”²*

(2) Discuss the agricultural biodiversity of Papua New Guinea?

Papua New Guineans settled on the land and became farmers, cultivating the land and utilizing the abundant biodiversity for food and feed over 9,000 years ago while the Europeans were still hunters and gatherers. The people in the Kuk Valley in Western Highlands Province started the global Agricultural Civilization where drainage, planting, digging and staking of plants have been used to cultivate taro, banana and yam between 7,000 – 10,000 years ago. These agricultural practices have propagated throughout different societies that settled in PNG dating back to 9,000 years. As a result, PNG has the world’s largest agricultural biodiversity and therefore the world’s largest agricultural gene pool.

As a result, Papua New Guinea has developed into a very rich agricultural biodiversity hotspot in the world, with farmers growing over 200 species of plants for food and feed. At present, of the country’s seven million people, 70% depend on agriculture.³



Figure BD-AB3: The link between ant agriculture and early social complexity (Adapted from: Jill Levine, 2016)

The country is also the world's centre of diversification of traditional root and tuber crops, leafy vegetables, fruits and nuts, cooking bananas, underutilized food crops, sugarcane, medicinal

² World Convention of Biodiversity, 2008.

³ <http://sdg.iisd.org/news/bioversity-international-promotes-agrodiversity-for-climate-resilience-in-papua-new-guinea/>

plants, ornamental trees and timber trees. These are a rich and valuable resource that must be appropriately utilised and conserved for future generations.⁴

Regarding agro-biodiversity, the sweet potato is a central component of the Papua New Guinean diet, with an estimated 5,000 cultivars of this staple found within the country. Numerous other plant species have traditionally been cultivated, including –

- more than 30 root crops,
- 21 legume species,
- 40 leafy green vegetables,
- 60 other vegetables and roots,
- 43 varieties of nuts,
- 102 fruits, and
- 89 other plants used for food or seasonings.

Wildlife plays an important part in traditional diets, supplying the primary intake of proteins and fats in many highland areas and other isolated areas of the country. In coastal areas, a wide variety of seafood, including fish, mollusks, and turtles, dominates local diets.⁵

(3) Discuss the wild relatives of agricultural crops in Papua New Guinea

Wild Relatives of Agricultural Crops

The crop wild relatives (CWR) of Papua New Guinea constitute an enormous and unique resource of genetic diversity which may be vital for future crop improvement, climate resilience and food security, particularly because they may harbor many valuable traits for agricultural adaptation to changing climate. Yet so little is known about the CWR in a country so biodiverse. Sadly, we know so little about their numbers, their distribution or those species or populations most at risk. All this while habitats in PNG continue to be lost or degraded.⁶

Kambuou et al recently searched for CWR in PNG and found several species of wild taro and wild rice. The location maps are shown below:

⁴ Kambuou, R.M. Protection of intellectual, biological and cultural property in Papua New Guinea, pp. 125 (<http://press-files.anu.edu.au/downloads/press/p229831/pdf/ch084.pdf>).

⁵ <https://www.cbd.int/countries/profile/?country=pg>

⁶ Kambuou, R.N., Okpul, T. and Hunter, D. (2014), Crop Wild Relative, 8: 39-42 (https://www.biodiversityinternational.org/fileadmin/user_upload/online_library/publications/pdfs/Papua_New_Guinea_a_much_neglected_hotspot_of_crop_wild_relative_diversity_1925.pdf)

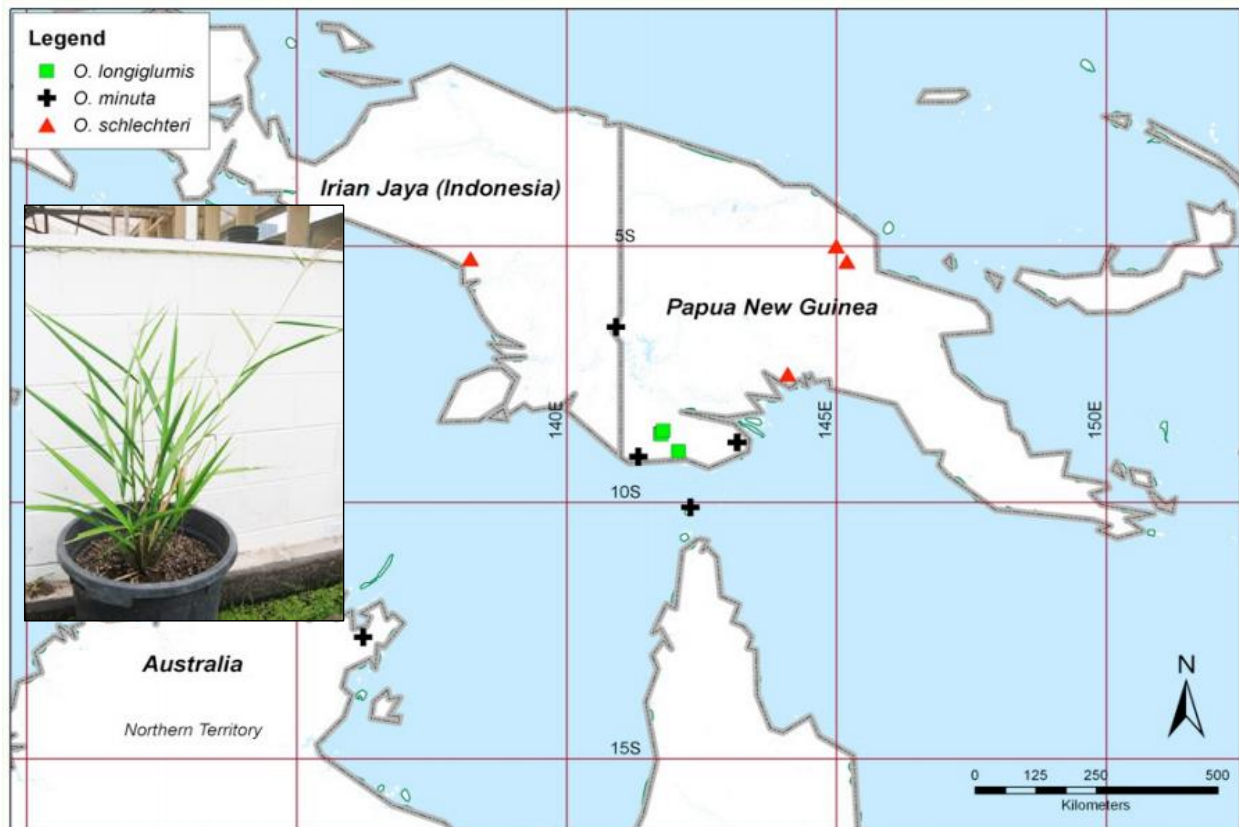


Figure BD-AB4: Distribution of high priority wild rice relatives in PNG (Source: Maxted and Kell, 2009).
insert: *Oryza ridleyi* species grown in a pot at UniTech (Photo: Prof. Tom Okpul).

(4) Describe the limitations to agricultural development in Papua New Guinea

Limitations to Agricultural Development in Papua New Guinea

Agricultural civilization in Papua New Guinea did not evolve into an advanced stage much like the Asian or the European agricultural development that includes large scale farming for two specific reasons.

1. The Volcanic Eruption of Mt. Hagen diverted the Waghi River towards Simbu Province. A high montane lake was consequently formed which flooded the Kuk Valley. This natural disaster has caused the early settlers to migrate up into the mountains and became more animistic (believe in a deity). The lake crested and cuts through narrow gorges in Simbu to escape. Only in the 1950s after the government drained the swamp and created settlements did people from other provinces resettle in the newly created blocks (such as Awiamb, Kondopina, Banz, etc.)
2. There was no beast of burden, e.g. horse, buffalo, donkey, camels, or such, to support and advance agricultural practice at that time and up until now. Agricultural practices were only utilizing human workers with rudimentary tools like digging sticks and paddles and were unable to advance to large scale farming. The Asians utilized water buffalos and the Europeans utilized horses to advance to large scale farming. This is probably the biggest setback in PNG's advance to large scale farming.

Topic 2: Genetic Diversity

When agricultural civilization started in Europe people started moving about, interbred and the agricultural biodiversity was reduced. However, in Papua New Guinea, it turned out opposite. Even after agricultural civilization started in Papua New Guinea, people still retained the agricultural biodiversity and even increased it. This phenomenon can be explained in terms of the Four Levels of Biodiversity already mentioned above.

The early settlers remained isolated from each other because of geographical constraints such as thick forest, fast flowing rivers, high mountains, swamps, language barriers, cultural barriers, tribal fights, and so on. Similarly, biological diversity were constrained by these barriers and evolved in isolation of each other. So within a small area, roughly less than 1% of the global land mass, the ecological diversity caused both species and functional diversities to emerge and thrive in an effort to co-exist within this small area or colonize an ecological niche. Ultimately, each of these diversities culminated in an abundant genetic diversity that we see today.

Recently, scientists from Sanger Institute (UK) studied the genetics of 381 Papua New Guineans from 85 different languages to ascertain whether languages and cultural differences have any bearing on genetic diversity.

The scientists looked at more than a million genetic positions in the genome of each individual, and compared them to investigate genetic similarities and differences. They found that groups of people speaking different languages were surprisingly genetically distinct from each other. So it can be inferred that species, ecological, and functional diversities underpin genetic diversity with the population.⁷ The same appears to be true for both flora and fauna as numerous studies of native agricultural crops attest.

- Taro
- Yam
- Sweet Potato
- Sugar cane
- Banana

Activity 1: Case Study

⁷ <http://www.sci-news.com/othersciences/anthropology/genetic-diversity-papuan-new-guinean-peoples-05238.html>

Title: The Geopolitics of Banana

The Story of the Fe'i Banana



Figure SB-BD-AB6: Fe'i banana (left) and Cavendish (right) (Adapted from: Promusa.com)

The Fe'i banana is native to Papua New Guinea and the Pacific. It has a characteristic feature amongst the different banana species. It has highly coloured sap, pink through to bright magenta and dark purple. The flowering and fruiting stem stands upright (rather than drooping), so that bunches of bananas are also upright. Ripe fruit has brilliant orange, copper-coloured or red skin with orange or yellow flesh inside. When eaten, the urine colour turns a bright orange. The banana has high vitamin A content.

Professor James Dale and his team of researchers from the Queensland University of Technology have cloned the gene coding for vitamin A from the Fe'i banana from Papua New Guinea and incorporated into the Cavendish banana to enhance its vitamin A content.

The genetically modified Cavendish banana was donated as a gift from Australia to the people of Uganda. This modified Cavendish banana is now growing in that African country.

This is an example of a gene cloned from a native Papua New Guinea banana has found its use in Biotechnology to solve vitamin A deficiencies in African countries. This highlights two key issues:

1. The need for Papua New Guinea to protect its abundant agricultural biodiversity by start building genomic database of these crops; and
2. Papua New Guinea's abundant biodiversity can be harnessed to address global biosecurity and climate resilience issues.



Figure SB-BD-AB7: Comparison of Karat Pwehu, one type of Fe'i banana, with white fleshed Utin Menihle, peeled and unpeeled.

(Credit: Bioversity International/L. Englberger)

Activity 1: Case Study

Title: Geopolitics of Banana

- The Story of the Fe'i Banana - The case study can be given to students to inspire them of the genetic diversity of Papua New Guinea that is being explored and utilized in other countries. The teacher can discuss the case study with students and motivate them to do more research and identify native agricultural crops in PNG that can be used for purposes of economic development.
- Why is our rich biodiversity not being legislated from foreign intrusion?
- Are there government agencies that protect our rich biodiversity as legislated?
- What is the capacity like at research institutions to discuss about the importance converting our biodiversity that are native into gene pools as genetic biodiversity?
- Discuss the economic value of our biodiversity as it is related to PNG's native Fe'i Banana
- Use the case study to discuss why PNG government should be careful or made aware of foreign policies that intrude into extraction PNG's biodiversity for their gain.

REFERENCES:

1. *Science News from research organizations: Golden bananas high in pro-vitamin A developed - Research has produced a golden-orange fleshed banana, rich in pro-vitamin A. Queensland University of Technology; July 7, 2017.*

Activity 2: Project

Title: Research, Case Write up and Presentation

- Research and discuss the wild relatives of agricultural crops native to Papua New Guinea. Given below are two examples identified.
 - Wild Taro
 - Wild Rice
- You are to do further research and do a presentation and write up for assessment, your teacher is to give.

MODULE 2: ORNITHOLOGY

Topic 1: Birds of Paradise

Activity 1: Classroom Activity

Title: Diversity of Birds in PNG

(1) Discuss the diversity of bird species within Papua New Guinea

Papua New Guinea has a rich biodiversity of birds. There are over 79 families and 730 species throughout the country, PNG. These birds are classified according to their ecological niche and migratory patterns and they are; breeding land and freshwater species, seabirds, migrants from the north, and migrants and wanderers from Australia and New Zealand. Papua New Guinea is amongst the eight Endemic Bird Areas around the World and caters for the largest diversity of about 320 endemic bird species. Some bird species in PNG are yet to be studied and classified.

The largest birds in Papua New Guinea are the flightless **cassowaries**, of which all three species are native to PNG. Two of these species: the southern cassowary (*Casuarius casuarius*) and the northern cassowary (*Casuarius unappendiculatus*) reach heights of 1.8 metres (6 ft). The cassowary is one of the world's most dangerous birds, for it is capable of inflicting fatal injuries with its powerful legs and the dagger-like claw on its inner toe. It is known to have killed humans.

The **pigeons** and **parrots** are well represented in Papua New Guinea. They achieve their greatest evolutionary diversity in PNG for the island is abundant in fruits and nectar producing plants. The parrots of PNG are very diverse with 46 species, a seventh of the world's total. The 45 species of pigeons, including the three crowned-pigeons are the largest pigeons in the world, and are a sixth of the world's total.

The **passerines** display the greatest amount of diversity with over 33 families within Papua New Guinea. The passerines of PNG are mostly small, often colourful birds which mostly inhabit the forested regions. The best-known family in PNG is the **Paradisaeidae**, one of three families there known collectively as **birds-of-paradise**. Many species show extravagant sexual dimorphism. The males can be ornamented with bright, iridescent colours, and modified, ornamental feathers such as tufts and wattles. They also display mating rituals, in which they undergo elaborate movements and calling, to attract females. Some species do not show sexual dimorphism; both male and female can have or lack ornamentation.

Closely related to the birds-of-paradise are the **bowerbirds**, a group of twenty rather drab, stocky and short-plumed birds found in Papua New Guinea. They lack the bright and iridescent color and ornamental plumes found in the birds-of-paradise, but the lack is compensated for with the male's *architectural skill*. The male builds and decorates an elaborate bower, ranging from mats, stick towers, avenued chambers to tipi-roofed huts and displays it to the females.

Another odd avifauna from Papua New Guinea is the poisonous birds, notably the **hooded pitohui**. In 1989, scientists discovered that the **hooded pitohui is poisonous**. Later, they

learned that the feathers and other organs of the pitohui were found to contain **batrachotoxin**. Since then, six other songbirds in PNG have been found to possess the same toxin in varying amounts.

Papua New Guinea's top predator is the New Guinea **harpy eagle** (*Harpyopsis novaeguineae*). PNG shares with the Philippines and New Zealand the distinction of having this bird as top predator.

Most seabirds' native to PNG is found throughout the tropics. Although some species are hunted for meat, valuable plumes and feathers, or for the pet trade, the main threats to most species come from logging and conversion of forest for agriculture, both of which degrade or eliminate important habitat.

The **birds-of-paradise** are members of the family **Paradisaeidae** of the order Passeriformes. The majority of species are found in Papua New Guinea. The family has 42 species in 15 genera. The members of this family are perhaps best known for the plumage of the males of the species (the majority) which are sexually dimorphic, in particular the very long elaborate feathers extending from the beak, wings, tail or head. For the most part they are confined to dense rainforest habitat. The diet of all species is dominated by fruit and to a lesser extent arthropods. The birds-of-paradise have a variety of breeding systems, ranging from monogamy to lek-type polygamy.

A number of species are threatened by hunting and habitat loss.



Figure SB-BD-O1: The Greater Bird of Paradise (Adapted from: Wikipedia)

(2) Identify the Scientific classification of the bird of paradise

Taxonomy of Birds of Paradise

Scientific classification

Kingdom: Animalia
Phylum: Chordata
Class: Aves
Order: Passeriformes
Superfamily: Corvoidea
Family: Paradisaeidae
(Vigors, 1825)

Genera: 15 genera, 42 species

genus: Lycocorax

- Halmahera paradise-crow, *Lycocorax pyrrhopterus*
- Obi paradise-crow, *Lycocorax obiensis*

genus: Manucodia

- Glossy-mantled manucode, *Manucodia ater*
- Jobi manucode, *Manucodia jobiensis*
- Crinkle-collared manucode, *Manucodia chalybatus*
- Curl-crested manucode, *Manucodia comrii*

genus: Phonygammus

- Trumpet manucode, *Phonygammus keraudrenii*

genus: Paradigalla

- Long-tailed paradigalla, *Paradigalla carunculata*
- Short-tailed paradigalla, *Paradigalla brevicauda*

genus: Astrapia

- Arfak astrapia, *Astrapia nigra*
- Splendid astrapia, *Astrapia splendidissima*
- Ribbon-tailed astrapia, *Astrapia mayeri*
- Stephanie's astrapia, *Astrapia stephaniae*
- Huon astrapia, *Astrapia rothschildi*

genus: Parotia

- Western parotia, *Parotia sefilata*
- Carola's parotia, *Parotia carolae*
- Bronze parotia, *Parotia berlepschi*
- Lawes's parotia, *Parotia lawesii*
- Eastern parotia, *Parotia helenae* (Disputed)
- Wahnes's parotia, *Parotia wahnesi*

genus: Pteridophora

- King of Saxony bird-of-paradise, *Pteridophora alberti*

genus: Lophorina

- Superb bird-of-paradise, *Lophorina superba*
- Vogelkop superb bird-of-paradise, *Lophorina niedda*
- Lesser superb bird-of-paradise, *Lophorina minor* (Disputed)

genus: Ptiloris

- Magnificent riflebird, *Ptiloris magnificus*
- Groewing riflebird, *Ptiloris intercedens*

- Paradise riflebird, *Ptiloris paradiseus*
- Victoria's riflebird, *Ptiloris victoriae*

genus: *Epimachus*

- Black sicklebill, *Epimachus fastosus*
- Brown sicklebill, *Epimachus meyeri*

genus: *Drepanornis*

- Black-billed sicklebill, *Drepanornis albertisi*
- Pale-billed sicklebill, *Drepanornis bruijnii*

genus: *Cicinnurus*

- Magnificent bird-of-paradise, *Cicinnurus magnificus*
- Wilson's bird-of-paradise, *Cicinnurus respublica*
- King bird-of-paradise, *Cicinnurus regius*

genus: *Semioptera*

- Standardwing bird-of-paradise, *Semioptera wallacii*

genus: *Seleucidis*

- Twelve-wired bird-of-paradise, *Seleucidis melanoleucus*

genus: *Paradisaea*

- Lesser bird-of-paradise, *Paradisaea minor*
- Greater bird-of-paradise, *Paradisaea apoda*
- Raggiana bird-of-paradise, *Paradisaea raggiana*
- Goldie's bird-of-paradise, *Paradisaea decora*
- Red bird-of-paradise, *Paradisaea rubra*
- Emperor bird-of-paradise, *Paradisaea guilielmi*

genus: *Paradisornis*

- Blue bird-of-paradise, *Paradisornis rudolphi*

Activity 2: Case Study

Title: Behaviour of Birds of Paradise

The Greater Bird of Paradise roost near the forest floor and climb up to the top branches early in the morning to display his plumage to attract the opposite sex in an intricate sexual selection. It is ironical that a majestic creation of the bird world which has perfected the evolutionary selection process would have two seemingly opposite behaviors – roosting and courtship. These extreme behaviors (between roosting and courtship) has become embedded within the psyche of our public service where policy misalignments abound and the leadership perched on high chairs of seeming comfort but roost in run down places and squatter settlements. In PNG bird behaviors have profound consequences and lessons to glean from.

The characteristic features of the birds of paradise and their cultural significance to different cultures in PNG. For instance the traditional attire of the highlands, Momase, New Islands and Southern resembles the behavior of the bird of paradise as we closely resemble our national bird. **Identify and discuss those provinces that resemble the behavior of the bird of paradise**

Attributes of the Bird	Traditional Attire	Significance
Dancing & Displaying	Simbu	
	Jiwaka	
	Morobe – Kumul	

Discuss the behavior and attributes of the birds of paradise imbedded in the fabrics of society and also the behavior of public service in corporate sectors.

Attributes of the Bird	False Impression/Public Service misalignment
The birds of paradise produce unique sounds and bear many bright colors, with uniquely with beautiful feathers	Pride and arrogance
Male birds are known for their extravagant courting displays to attract female	Attract investors without grounding with sustainable public policies
The birds of paradise generally are comfortable in the upper ranges of the forest canopy and drops to sleep at night.	Public Service is comfortable in sectors with fame in positions that leads to dropping in vital service delivery.

Topic 2: Inspirations from Bird Behaviour

Activity 1: Class Activity

Title: Brood Parasitism

(1) Describe what brood parasitism is?

Brood parasitism happens in avian world when birds of a different species (parasites) manipulates another bird species (hosts) to raise its young as if it were its own (brood parasitism) or lay their eggs in another bird's nest (egg parasitism). In the case of the egg parasitism the parasitic bird lays its egg in the nest of the host and is incubated by the host until it hatches. The host then rears the parasitic hatchling until it takes flight. Few weeks after it takes flight the fledgling rejoins with its parasitic parents.

The best studied example of brood parasitism in birds is that of the cuckoo and the cowbird (both are parasites).



Figure 1: A shiny cowbird chick (left) being fed by a rufous-collared sparrow.

In PNG, especially up in the Highlands, a similar brood parasitism has been observed. Since there is no proper reference that is available for the study of the birds, the local names of the birds in Kuman language of Simbu is used. The parasitic bird is called “mendondi” and the host is “tenduage”. Mendondi is a mid-size brown/grey color and utilizes the nests of tenduage – a small black and white bird with colors resembling the Willie wagtail (*Rhipidura leucophrys*) or the New Zealand Tomtit shown in the photo below.



Figure 2: A NZ Tomtit (Photo: Copyright: © Shellie Evans)
(Source: <https://images.app.goo.gl/hz4cpPyg3GvPt4ZJ9>)

Implications:

Brood parasitism is a tool applied in geopolitical and corporate boardroom predatory behaviour. Two cases are presented here to highlight this avian behavior.

Activity 2: Case Study

Title: National Agriculture Research Institute (NARI)

The National Agriculture Research Institute was established in July 1996 by an Act of Parliament. Its purpose was to undertake research in agriculture, specifically rice so that PNG can become self-sufficient in rice production. However, after the institute was established its funders directed its research mostly towards taro (*Colocasia esculenta*). Several Masters and Doctorate degrees were awarded for research into taro. In the end it was realized that taro cannot be exported as a commodity because it was endemic to PNG and any export of the crop would result in the export of pests and pathogens as well.

The government of PNG is an example of the Tenduage who built the nest (NARI) for the Australians (Mendondi) to lay their eggs (their interest). This has misled the original intent to produce rice in PNG which was a diversion that is implied in brood parasitism. Hence, the government of PNG is now fully funding the rice project in Taiwan that should have been grown in PNG. However, all the Taiwan rice are packaged in Australia and imported again to PNG.

Questions

1. How can the PNG government identify and assess foreign interests that intrude right into our established institutions and redirect the country's development agendas?

2.

Activity 3: Case Study

Title: Bemobile

Bmobile is a State-owned mobile telephone company. It was operating since 1997. Prior to that it existed as a mobile phone business unit within Telikom PNG. The company was owned 50% by the State of PNG, 35% by GEMS of Hong Kong and 5% by NASFUND, SuperFund and ADB. GEMS in turn was owned by Capital Way of Cayman Islands and Capital Way International of British Virgin Islands.

According to the company's constitution, there were 7 board members. Although, GEMS was a minority shareholder it was entitled to nominate 4 Board members and provide management. So effectively the control of the majority PNG-owned company was outsourced to its minority overseas partner. Bemobile rolled out CDMA mobile phone network which was an obsolete technology while its competitor was rolling out GSM technology.⁸ There were other cases of product and customer mismatch. In the end Bemobile lost its significant market share from as high as 80% to a measly 3% market share almost a decade later.

Therefore, Digicel is now a monopoly in the communications and technology market in PNG. This is a classical board room predatory behavior inspired from brood parasitism.

Questions

1. How can the PNG government identify and assess foreign interests that intrude right into our established corporate board rooms and redirect the country's development agendas?

2.

Activity 4: Project

Title: Research

After going through the case studies. Students should be developing this curiosity to find out more. Thus, research projects and assessments should be given for students to identify and do case studies on bird species of interest in PNG. This should cover the two topics:

1. The Birds of Paradise and its national and cultural significance
2. Inspirations from bird behaviours identified within their society:
For example, DukDuk is a custom practice amongst the Tolai's with inspiration gleaned from the Guria Pidgeon (Goura Victoria – Victoria Crowned Pidgeon).

You will be able to interpret and discuss the case studies and relate to or recognize similar real life case scenarios. Another example would be the Israel Warfare in the Middle-East involving the USA. This might be research topic or other similar activities in relation to inspirations from ornithology or other inspirations from nature.

⁸ Difference between CDMA and GSM technology: CDMA mobile phones come with no sim card. Number are fixed to the phone. GSM has the flexibility of using a sim card so it is easier to remove just the sim and use it on another phone.

Application of Knowledge, Skills and Values (AKSA)

1. The knowledge gained in biodiversity would encourage students to study the agricultural biodiversity and how to utilize its benefits.
2. Students would develop a keen interest in exploring Papua New Guinea's biodiversity.

References:

1. Bolger J. Sciencing: four types of biodiversity. Leaf Group Ltd. March 09, 2018. (Accessed online 18-02-21: <https://sciencing.com/difference-between-community-ecosystem-5776576.html>)
2. Convention on biological diversity. What is Agricultural Biodiversity? UN environment programme. (Accessed online 15-02-21: <https://www.cbd.int/agro>)

THEMA BIOTECHNOLOGY

MODULE 1: CONCEPTS AND APPLICATIONS OF BIOTECHNOLOGY

Topic 1: Recombinant DNA Technology

Activity 1: Classroom Activity

Title: Questions and Answer

(1) Define and describe biotechnology

Biotechnology combines biology with technology to engineer or use, modify or upgrade the part or whole of biological system for industrial and human welfare. It combines various technologies derived from biology, mathematics, physics, chemistry and engineering to modify the biological systems. Its application ranges from agriculture (Animal Husbandry, Cropping system, Soil science and Soil Conservation, Plant Physiology, Seed Technology etc and Crop Management) to industry (food, pharmaceutical, chemical, byproducts, textiles etc.), medicine, nutrition, environmental conservation, cell biology, and so on. Biotechnology is now one of the fastest growing fields of sciences.

(3) Define and describe Recombinant DNA technology

Recombinant DNA technology allows for the fragments of DNA molecules from two or more species that are recombined and then inserted into a host organism to produce new genetic combinations that are useful to science, medicine, agriculture, and industry. This technology makes it easy to isolate a particular gene that is of interest to researchers so that they can determine its nucleotide sequence, study its transcripts, mutate it in highly specific ways, and reinsert the modified sequence into a living organism to produce the corresponding protein.

(4) Define and describe DNA cloning

DNA Cloning – Plasmids - A clone is a cluster of individual cells that are genetically identical because they descended from one progenitor (common ancestor). Every time cells replicate they produce identical daughter cells.

In recombinant DNA technology, scientists are able to produce many identical copies (or clones) of a single fragment of DNA (known as a gene). In practice the procedure is carried out by inserting a DNA fragment into a replicating molecule called a DNA cloning vector (carrier). The most commonly used vectors are plasmids (circular DNA molecules that originated from bacteria), viruses, and yeast cells. Plasmids are useful for bacteria's survival but are not the entire genome of the bacteria. They are small enough to be conveniently manipulated for DNA cloning.⁹

The plasmids are then inserted into the bacterial cells and multiple copies or clones are made. In addition, as the bacterial cells multiply the daughter cells make exact copies of the progenitor. After the clones are produced the bacterial cells are destroyed to recover the plasmids.

⁹ <https://www.britannica.com/science/recombinant-DNA-technology>

(5) Define and describe Restriction Sites and Restriction Enzymes

Restriction Sites and Restriction Enzymes - There are specific sites on the DNA sequence called restriction sites that are recognized and cut by specific enzymes called restriction enzymes. These restriction enzymes act like “molecular scissors,” cutting the DNA into specific target sequences. The most suitable restriction enzymes cut at the restriction sites of both the donor DNA and the vector DNA and leave one strand of the DNA unpaired and hangs over the other strand. In the ligation (recombination) stage these unpaired nucleotides (called “overhangs”) of the donor DNA and vector DNA are paired because of their complementary overhangs resulting in a recombined DNA molecule known as a recombinant DNA.

(6) Define and describe DNA Amplification with Polymerase Chain Reaction

DNA Amplification – Polymerase Chain Reaction - PCR method can be used to amplify the gene (DNA fragment) of interest using primers with specific restriction sites. Primers are short complementary single stranded DNA molecule that binds on either the 5' or 3' ends of the DNA molecule and assembles the complementary DNA strand. The PCR process is explained in the Chemistry Phylum. PCR can be used to amplify the gene of interest from the recovered plasmids. Alternatively, the gene (DNA fragment) of interest can be excised from the recovered plasmids with the restriction enzymes.

(7) Define and describe Recombinant Expression and Production of Proteins

Recombinant Expression – Production of Proteins - The expression vector (a different plasmid) is treated with restriction enzymes to open the site for the gene of interest to be inserted. The recombinant plasmid, after the gene of interest is incorporated, is then inserted into the bacteria and the grown on culture medium to select the colonies that contain the gene of interest. The selection process to ensure the colonies with the gene of interest are selected uses antibacterial agents. The colonies are then grown in large broths to produce the corresponding recombinant protein from the gene of interest.

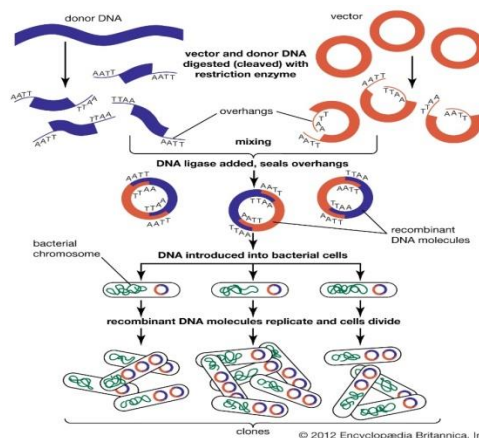


Figure SB-BT1: Steps involved in the engineering of a recombinant DNA molecule.

Topic 2: Industrial Biotechnology

Activity 1: Classroom Activity

Title: Industrial Biotechnology

(1) Describe what genetically modified or engineered organisms are?

Some of the applications of recombinant DNA technology are applied in industrial biotechnology. Some of these are listed below:

Genetically Engineered Organisms

The recombinant DNA technology has made it possible to add the DNA of one organism to the genome of another. The added gene is called a transgene. The transgene is inserted into a chromosome of the target organism and is passed to the progeny as a new component of the genome. The resulting organism carrying the transgene is called a transgenic organism or a genetically modified organism (GMO). In this way, a “designer organism” is made that contains some specific change required for an experiment in basic genetics or for improvement of some commercial traits.

In agricultural biotechnology different desired traits, such as insect resistance and disease tolerance genes, have been inserted into plants through recombinant DNA technology. Transgenic plants, such as corn and cotton, have been produced. Genes for toxins that kill insects have been introduced in several transgenic species, including corn, soybean and cotton. Bacterial genes that confer resistance to herbicides also have been introduced into crop plants.

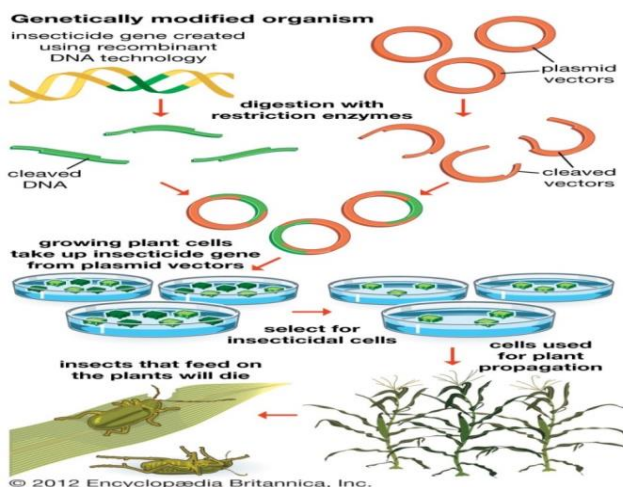


Figure SB-BT2: Genetically modified organisms are produced using scientific methods that include recombinant DNA technology.

(2) Discuss the applications of recombinant DNA technology in medical or pharmaceutical (health care) and agricultural industries

Gene Therapy

In gene therapy defective genes that are responsible for genetic diseases can be “fixed” by introducing a normal gene into an individual’s genome in order to repair the defective mutation that causes a genetic disease.

Diagnostics

Recombinant DNA technology has led to powerful diagnostic procedures useful in both medicine and forensics. In medicine these diagnostic procedures are used in counseling prospective parents as to the likelihood of having a child with a particular disease, and they are also used in the prenatal prediction of genetic disease in the fetus.

In forensics, DNA fragments called variable number tandem repeats (VNTRs), which are highly variable between individuals, are employed to produce what is called a “DNA fingerprint.” A DNA fingerprint can be used to determine if blood or other body fluids left at the scene of a crime belongs to a suspect.

Genomics

The genetic analysis of entire genomes is called genomics. Such a broad scale analysis has been made possible by the development of recombinant DNA technology. In humans, knowledge of the entire genome sequence has facilitated searching for genes that produce hereditary diseases. It is also capable of revealing a set of proteins—produced at specific times, in specific tissues, or in specific diseases—that might be targets for therapeutic drugs.

Protein Manufacture

Recombinant DNA procedures have been used to convert bacteria into “factories” for the synthesis of foreign proteins. This technique is useful not only for preparing large amounts of protein for basic research but also for producing valuable proteins for medical use. For example, the genes for human proteins such as growth hormone, insulin, and blood-clotting factor can be commercially manufactured. Another approach to producing proteins via recombinant DNA technology is to introduce the desired gene into the genome of an animal, engineered in such a way that the protein is secreted in the animal’s milk, facilitating harvesting.

Activity 2: Project/Case Studies

Title: Application of Biotechnology

This research project can be optional for teachers

1. Case Studies of Genetically Modified Crops and Organisms
2. Case study of Synthesis of Glucose From Lignin (Woods) by Wood Worms¹⁰

Application of Knowledge, Skills and Values (AKSA)

1. The knowledge gained in Biotechnology would globalize the mindset of students in terms of its applications.
2. Students would think like global citizens in their own country and how to utilize technology to improve standard of living and sustain livelihood.

¹⁰ <https://www.popularmechanics.com/science/energy/a25399474/gribble-worm-renewable-energy/>

THEMA

HEALTH AND

MEDICINE

MODULE 1: HEALTH AND MEDICAL BIOLOGY

Topic 1: Pharmacology

Activity 1: Classroom Activity

Title: Questions and Answer

(1) Introduction to pharmacology

Modern medicine relies heavily on drugs as the main tool of therapeutics (i.e. mostly treatment of disease rather than cure). The study of **drugs** or **chemicals** and the effects they have on living animals is called *pharmacology*. It explains what drugs are, what they do to body systems and functions and what the body does to the drugs and furthermore elimination of it from the body. It also explains why a person may experience side effects when they take drugs and why there is such a wide spectrum of differences between drug actions in different people.

Everyone at one stage or another in life will rely on a **pharmaceutical product**, whether it is for themselves, a friend or a family member. Therefore it is useful to have a basic understanding of pharmacology which is discussed in this topic.¹¹

(2) What is a drug?

A drug is a chemical that interacts with proteins in the body to affect a physiological function. This is the general idea behind all **medicine**. Once these chemicals are absorbed into the systemic circulation they bind with certain proteins and this changes the functioning of the cell slightly. For example, anticancer drugs bind to proteins on the surface of cancer cells this stimulates the cells to die. In this case cell death is the physiological action of the drug.

No drugs are specific to interacting with just one type of cell or one type of protein and this is what causes side effects. Again using an anticancer drug as an example, the medication works by binding to very rapidly dividing cells, such as cancer cells, however hair cells are also rapidly dividing and that is why one of the side effects of anticancer drugs is hair loss.

(3) What are the contents of a drug?

The chemical in the drug that affects physiological functioning is the *active ingredient* of the drug. For most drugs, the amount of chemical needed to cause an effect is very small, often as small as 5 micrograms; this is 0.005% of a gram! As you can imagine this is too small to package and handle, these ingredients are very expensive and giving out little amounts like that will cause most of the drug to be lost and wasted. Therefore most of the drugs that we take are also comprised of inactive ingredients that work to fill out the drug. Inactive ingredients are, as the name suggests, ingredients that have no effect on the functioning of cells, namely *lactose*, *dyes* and *gluten*. If the drug needs to be taken orally, the inactive ingredients also work to bind the drug together and lubricate the drug so it is easy to swallow.

So the inactive ingredients are the fillers, binders and lubricants of the drug whereas the active ingredient is the very small amount of chemical that reacts with the body to produce an effect.

¹¹ What are drugs? An introduction to pharmacology (30 Jul 2009)¹¹

<https://healthengine.com.au/info/what-are-drugs-introduction-to-pharmacology#:~:text=A%20drug%20is%20a%20chemical,functioning%20of%20the%20cell%20slightly>].

(4) How do drugs work?

Our bodies are largely controlled by proteins. Proteins exist in many different forms in the body and have many different functions. Each protein has a specific function and is quite specific to the cell type that it acts on. For example, there are specific types of proteins called receptors. Receptors are embedded on the cell surfaces, there are different receptors for different types of cells. A liver cell will have different receptors than a cardiac cell. The receptor binds to other proteins and chemicals on the outside of the cell and this in turn creates a change in the functioning of the cell. Proteins also act as drug targets. In order for a drug to exert an effect it needs to be bound to a protein. This can be thought of as a lock and key system; where the drugs are the key and the protein is the lock. Once the drug is bound in this lock and key mechanism it can have one of two main influences over the cell. It can produce a change in response or it can stop a normal response of the cell. Drugs that produce a change in the cell functioning are called *agonists*. Drugs that stop a normal function of the cell are called *antagonists*.

(5) Pharmacodynamics: What the drugs does to the body?

Once the drug is bound to a protein it exerts a **therapeutic effect** on the body, this is the *pharmacodynamics* of a drug. There is an enormous list of different drugs and their actions in the body.

(6) Pharmacokinetics: What the body does to the drugs?

Pharmacokinetics is the study of what happens to drugs once they enter the body. The main stages include:

- The **absorption** of the drug into the blood and across *cell membranes* to enter the cells;
- The **distribution** of the drug throughout the body;
- The **metabolism** or breakdown of the drug; and
- The **excretion** of the drug from the body.

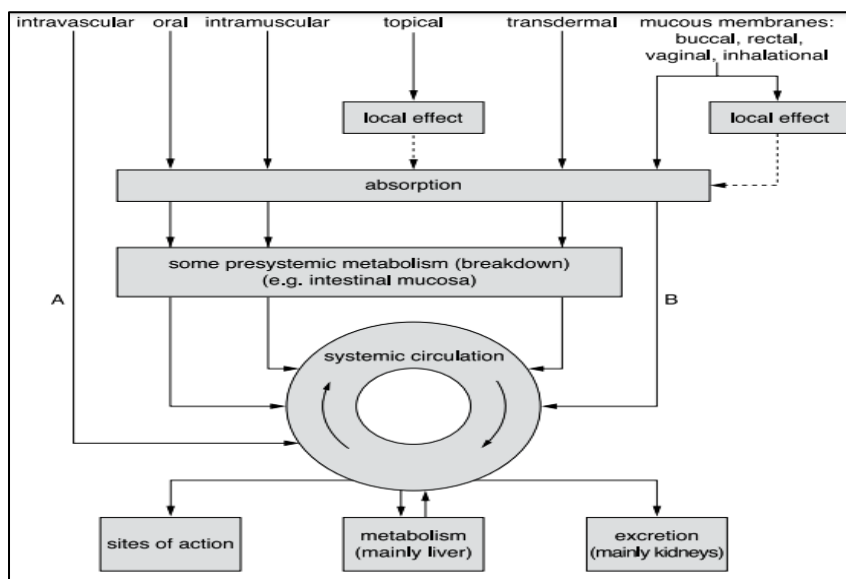


Figure HM-PC1: Summary flowchart of drug absorption, distribution, metabolism and excretion. (Adapted from: Pharmacokinetics and Pharmacodynamics - Medicines and the Body, 2021)

Each drug will have a unique *bioavailability*. This is the amount of drug *available* to have an effect on the biological system. A drug's bioavailability is determined by its pharmacokinetics. For example, some drugs are poorly absorbed as they do not cross cell membranes as quickly or as effectively as others and so less of the drug will pass into the systemic circulation where it needs to be in order to have an effect.

The proportion of the drug that does pass into the circulation is called the *drug-plasma concentration*. When a drug is absorbed into the circulation, the plasma concentration will increase until it reaches a peak and then as the drug is metabolised this plasma concentration will decline until the entire drug has been metabolised and then excreted from the body. Depending on the characteristics of the drug some will reach the peak plasma concentration quicker than others or be metabolised faster and so on.

Each drug has a range of dosages that can effectively treat a condition while still remaining safe. That is, the range between the lowest dose that has a positive effect, and the highest dose before the negative effects outweigh the positive effects. This is known as the *therapeutic window* of the drug. This can vary substantially between different types of drugs. For example, one drug could be safe and efficacious anywhere between 5mg to 20mg whereas another could have the therapeutic window between 15mg and 20mg.

(7) Why do the effects of drugs vary between different people?

Many of the reasons that we see such a wide and diverse range of efficacies of drugs across people are that drugs work differently in different people. A drug will usually produce the same qualitative effect across individuals, that is to say that it will produce the same end result and the same side effects but the quantity of these effects will be different. So some people may experience a shorter action of the drug or a more intense side effect.

This variation is due mostly to differences in pharmacokinetics and pharmacodynamics between ethnicity, age, genetic makeup and disease state.

Ethnicity

There are quite substantial differences in drug metabolism between people of different ethnicities. Asians are usually more sensitive to most drugs than Caucasians and Caucasians are more sensitive than afro-Caribbean's.

Age

Elimination of the drug from the body is directly influenced by age. Newborns and elderly experience the effects of drugs for longer and the drug takes a lot longer to be eliminated from the body.

Newborns

When babies are born to term, their renal function is very quick to establish similar levels to adults within one week after birth. If the baby is born prematurely it can take 8 weeks or more to reach the level of enzymes necessary. If drugs are given before the renal function is at this level, the drug elimination from the body takes a lot longer and so do the effects of the drugs.

Elderly

Renal filtration rate begins to decline at 20 years of age and by 50 years of age it has declined by 50%. This again will affect the elimination of drugs from the body.

Genes

The differences in our genes are also an important determinant of variability in what our bodies do to the drugs.

Disease state

There are many different disease states that affect pharmacokinetics. In fact most diseases will affect pharmacokinetics to some degree and this is for your doctor to determine and consider when prescribing medication. Diseases of the liver and kidneys will affect drug metabolism and excretion whereas diseases of the gastroenterological systems will affect the absorption of drugs. Receptors, the blood-brain barrier, blood, heart and skin are just some other areas that, if affected by disease, can impair the therapeutic action of drugs.

(8) Discuss the importance of drug selection during pregnancy

During pregnancy the process of drug elimination from the body becomes very complicated. The increased cardiac stress put on the mother's body enables the kidneys to increase their filtration rate. This is the *opposite* for the growing baby; the drugs that cross the placenta to the foetus will be eliminated very slowly as the kidneys and liver are not fully developed. Taking medication anytime during pregnancy can affect the growing foetus but the first trimester is the high risk zone.

As a result, when a woman is pregnant their doctor has an especially difficult job in weighing up the risks versus benefits of prescribing a drug. Not only must the patient's risk benefit be determined but also the risk for the developing foetus. Determining foetal risk from drug exposure is near impossible due to the lack of scientific data in the area.

That said, there is a list of drugs that are known to be harmful to a developing baby and these are called **teratogens**. Examples of these are alcohol and cigarette smoke. Teratogens **must be strictly avoided** during pregnancy. If you are in doubt as to whether a medication you are taking could be teratogenic always check with a doctor.

When weighing up the risks of prescribing medication the therapeutic decision is not the sole responsibility of the doctor but also the mother. Mothers will be made aware of the *possible* risks of the treatment and, through discussions with a doctor and family can decide for themselves whether they want to take the medication.

As a guide all drugs have been assigned a pregnancy category. The pregnancy categories describe the relative risk associated with the medication by summarising the information that is available to date. Each category puts into perspective the potential reliability of the information by describing the number of women that have been pregnant or of childbearing age that have taken the medication and the observable effects on the developing foetus. The categories provide an invaluable tool for weighing up the risk-benefit for doctors and patients.

(9) Discuss the importance of why mothers must consult doctors about the drugs prescribed while breastfeeding

Breastfeeding can expose a feeding child to toxicity due to the medication the mother is taking. The relative risk of toxicity through breast milk can be estimated by a doctor by weighing up the dose of the medication given, the amount from this that could be potentially excreted in breast milk and finally the potential dosage the infant will absorb systemically from the milk they ingest. Similar to the issue of medicating during pregnancy, large clinical trials have not been conducted in breastfeeding mothers and therefore the evidence is not available to support any **certainty** that the child will not be affected. If the mother needs to take medication which is known to be risky for children then the mother should not breastfeed her child.

(10) Discuss the advantages and disadvantages of Over-the-counter (OTC) medication

Most pharmaceutical products cannot be purchased without a prescription from a doctor. That said, there are quite a number of non-prescription or over-the-counter (OTC) drugs available that do not need clearance from a doctor in order to purchase and use. Depending on the particular medication, some OTC medicines can be found in supermarkets, convenience stores

and petrol stations or can be purchased online. Buying medicines over the counter can be advantageous in that it allows the consumer to self-medicate minor health problems giving more of the power and control of health conditions to the individual.

There are disadvantages involved with non-prescription medications. As OTC drugs do not require a doctor's visit they have the potential to mask more serious medical issues. OTC drugs can also be associated with misuse and dependence.

Dependence on OTC medication is prevalent in society. The majority of consumers who do become dependent initially purchase OTC medication with the intention of treating a minor health issue but without proper guidance find it difficult to cease use of the drug. Only a small proportion of consumers purchase OTC with the intention to misuse the medication.

Before purchasing any over the counter medication it is advised that you speak to your pharmacist about the benefits, risks and appropriate use of the medication. Therefore it is wise that you purchase the medication from a community pharmacy rather than a supermarket.

e-pharmacies are online pharmacies where only non-prescription drugs can be purchased and delivered. Although this may seem like an easy option, a consumer cannot be sure they are receiving all the information they need in order to make an informed decision about whether to take the drug or whether further information is required. While some internet sites provide very accurate and comprehensive information there are some that provide poor quality information and do not identify possible drug interactions. Most OTC drugs are safe to use when used properly however it is always best to speak to your pharmacist or trained pharmacy assistant about dosing, drug interactions and other options.

An important point to note is that medications need to be suited to the individual, this is the area that pharmacists and doctors are trained to assess. When in doubt always ask a trained healthcare professional.

(11) Describe what drug–drug interactions mean

Polypharmacy is the use of multiple drugs at once for different reasons. This occurs most frequently in the elderly who need to take many different types of drugs a day. Drugs can be affected by the pharmacodynamics or pharmacokinetics of the other drug. For example a pharmacodynamic reaction would be if drug X works to decrease heart rate it may affect the distribution of drug Y due to the decrease in blood pressure. A pharmacokinetic reaction would be if both drugs are metabolised in the liver and therefore need to compete with the metabolism enzymes.

(12) What are the risks and benefits of drugs

The side effects associated with some medications are more detrimental than the disease state they are treating. It is in these situations when you and your doctor must work together to determine the risk and benefit profile of taking the medication. In order to completely determine the risks you **MUST** tell your doctor which medications you are taking concomitantly, including herbal supplements, any allergies you have and any previous adverse effects you have experienced from medication.

Make sure you are aware of all the medications you are taking, what they are for and the risks associated with each. If the side effects are severe you may not want to continue with the treatment. Ask your doctor if there are any alternatives and if not discuss the consequences of not taking the medication. However, never stop a medication without the advice of your doctor.

(13) Describe the stages in developing a pharmaceutical product

Developing pharmaceutical products is an extremely expensive and long process, costing billions of dollars and taking over a decade to produce one drug! There are three main stages in the drug development process. These are:

- **Laboratory methods:** this includes testing the newly formed/discovered molecule on cell cultures (these are cells in dishes in labs not within living organisms) and then animals (usually either rats or mice).
- **Clinical trials:** firstly on healthy human volunteers then moving onto patients undergoing medical treatment.
- **Socioeconomic methods:** assessment of the drug in the community, the adverse effects, the family of the patient's response, the healthcare costs etc.

Each stage of the process must pass strict safety and efficacy testing before the medicine can progress. Throughout all stages the scientists are determining the benefits versus risks of the new drug. For every drug there will be side effects and adverse effects in some patients. In order to determine whether the drug will have an overall benefit, the proportion of patients that respond positively to the treatment must be compared to the number of patients that respond negatively. So for example in a clinical trial sample of 1000 patients, if 996 patients experience a significant improvement in quality of life and 4 patients experience an adverse drug reaction, more than likely the benefit outweighs the risk in this case. If 550 patients significantly improve and 450 patients experience adverse drug reactions the drug is not likely to be marketed. This of course depends on the nature and severity of the adverse events.

Once the drug has been determined to be safe, effective and a significant cost-benefit it may be produced marketed and distributed to the public. In Australia, the authority that makes the overruling decision is the Therapeutic Goods Administration (TGA).

Topic 2: Immunization and Vaccines

Immunization with vaccines is a key component of primary health care globally. It saves millions of lives every year from previously untreated and unrestrained infectious diseases such as diphtheria, tetanus, pertussis, influenza and measles. Vaccination is critical to the prevention and control of infectious-disease outbreaks.¹² Currently, COVID-19 pandemic is a challenge for the pharmaceutical companies to create vaccines to combat SARS-CoV-2. Hence, immunization with vaccines underpins global health security and will be a vital tool in the battle against antimicrobial resistance. However, with the wrong intentions (bio-warfare) vaccination can be devastating.

Define and differentiate between immunization and vaccination

Immunization is the process whereby a person is made immune or resistant to an infectious disease typically by administration of a vaccine.¹³ It is the body's ability to generate an immune response naturally, or through vaccines that it builds immunity or resistance towards a particular disease. These vaccines are artificially synthesized and manufactured *antigenic* material that is administered to develop adaptive immunity to an infectious disease.

Table HM-IV1: Difference Between Vaccination And Immunization	
Vaccination	Immunization
The process involves Introducing a weakened / deactivated disease causing microbes into a person	The process starts after the person is exposed to the vaccine and the body starts building resistance to that disease
It is usually injected or administered orally	It is not administered in any way, the body develops the resistance from vaccines
Imovax Rabies is the trade name for rabies vaccine	The body builds up immunity through this vaccine for the disease rabies
Vaccination does not guarantee complete resistance to a disease	Complete immunity occurs when the person fully recovers from the disease
Usually, if mutation happens to microbe, it might render the vaccine ineffective (this is the reason why common cold has no vaccine)	Similarly, variations of a disease impact the body's ability to generate an immune response

(Adapted from: Difference between Vaccination and Immunization, BYJU'S)

What is the significance of Adaptive immune system and humoral immunity of the human body?

The adaptive (acquired) immune system provides a tailored response to each stimulus by learning to recognize molecules it has previously encountered. Adaptive (or acquired) immunity creates an immunological memory leading to an enhanced response to subsequent encounters with that same pathogen. This process of humoral response and acquired immunity is the basis of *vaccination*.

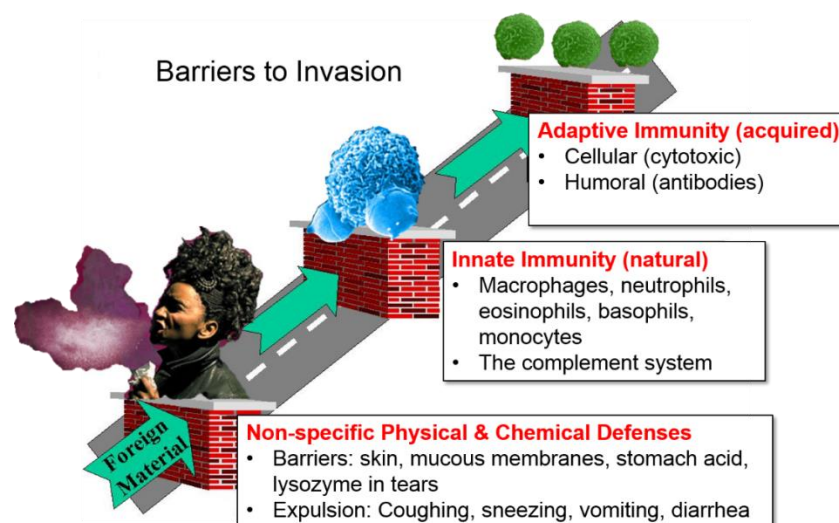
Describe the humoral immune response to an infectious disease

¹² WHO; vaccines and immunization (https://www.who.int/health-topics/vaccines-and-immunization#tab=tab_1)

¹³ <https://byjus.com/biology/difference-between-vaccination-and-immunization/>

The humoral response (or antibody-mediated response) involves B cells that recognize antigens or pathogens that are circulating in the lymph or blood. A major task of mature B-lymphocytes, also known as plasma cells, is to produce antibodies. These are tiny protein molecules that stick to pathogens, thereby turning these into recognizable enemies to be engulfed, digested, or killed by so called *guzzle cells* (macrophages) or *natural killer cells* (special T-lymphocytes). The response following this chain of events is described below:

1. The digestion of viral particles or bacterial proteins by Macrophages leaves behind parts of the invading germs called *antigens*. The body identifies antigens as dangerous and stimulates antibodies to attack them.
2. The B-lymphocytes produce antibodies that attack the antigens left behind by the macrophages.
3. The interleukins or helper T cells costimulate B cells. In most cases, both an antigen and a costimulator are required to activate a B cell and initiate B cell proliferation.
4. B cells proliferate and produce plasma cells. The plasma cells bear antibodies with the identical antigen specificity as the antigen receptors of the activated B cells. The antibodies are released and circulate through the body, binding to antigens. Antigens bind to Antibodies forming an antigen-antibody complex.
5. T cells also produce memory cells. Memory cells provide future immunity.



Describe How Vaccines Work

An immunizing vaccine consists of killed bacterial cells, inactivated viruses, and killed parasites, non-virulent forms of live bacteria, denatured bacterial toxins, or recombinant protein. The introduction of a vaccine into a human leads to protection against a virulent form of the micro-organisms or toxic agents that contain the same antigen. Antigens in non-virulent material cause differentiation of lymphoid cells (B and T lymphocytes) so that they produce antibody toward the foreign antigen and differentiation of some lymphoid cells into memory cells. Memory cells do not secrete antibody but place antibodies to the antigen onto their outer surface, where they act as future sensors for the antigen. These memory cells are like longstanding radar for the potentially virulent antigen. On reintroduction of the antigen later, binding of the antigen to the cell surface antibody in the memory cells stimulates the memory cell to divide into antibody-producing cells and new memory cells. Once introduction of an antigen occurs, this reduces the

time required for antibody production and increases the initial concentrations of antigen-specific antibody produced. This is the basis for the protection provided by immunization.

Identify the different types of vaccines

Scientists take many approaches to developing vaccines. These approaches are based on information about the infections (caused by viruses or bacteria) the vaccine will prevent, such as how germs infect cells and how the immune system responds to it. Practical considerations, such as regions of the world where the vaccine would be used, are also important because the strain of a virus and environmental conditions, such as temperature and risk of exposure, may be different across the globe. The vaccine delivery options available may also differ geographically.

There are several types of vaccines, including:

- Inactivated vaccines
- Live-attenuated vaccines
- Subunit, recombinant, polysaccharide, and conjugate vaccines
- Toxoid vaccines
- Viral vector vaccines
- Messenger RNA (mRNA) vaccines

Describe the different types of vaccines

1. Inactivated vaccines

Inactivated vaccines are dead or the killed version of the microbe (viruses, bacteria, etc.) that causes a disease or infection. The inactivated vaccines provide temporary immunity for a shorter period of time. Therefore, an individual needs several doses known as booster shots over time in order to get ongoing immunity against diseases.

Inactivated vaccines are used to protect against:

- Hepatitis A
- Flu (1 shot only)
- Polio (1 shot only)
- Rabies

2. Live-attenuated vaccines

Live vaccines use a weakened (or attenuated) form of the microbe that causes the disease. These vaccines create a strong and long-lasting immune response as they are synonymous with natural infection and immunity. Therefore, just 1 or 2 doses of most live vaccines can give you a lifetime of protection against an infectious organism and the disease it causes.

However, there are several disadvantages and limitations to the use of live vaccines. This are explained below:

- Live vaccines are contraindicated in individuals with weak immune systems as in organ transplants and chronic diseases or long term health illnesses. As they contain a minimum amount of weakened live viruses.

- Live vaccines have to be refrigerated or stored in refrigerators at all times to keep it cool. Therefore, they can't be used in countries with limited access to refrigerators.

Live vaccines are used to protect against:

- Measles, mumps, rubella (MMR combined vaccine)
- Rotavirus
- Smallpox
- Chickenpox
- Yellow fever

3. Subunit, recombinant, polysaccharide, and conjugate vaccines

Subunit, recombinant, polysaccharide, and conjugate vaccines are identified by their targeted use on specific key components of the microbe as in its molecular composition. Due to their specificity they give a very strong immunity, however booster shots are needed for further immunity against diseases as it lasts for a certain period of time and wears out. Another advantage is that they can also be used on almost everyone who needs them, including people with weakened immune systems and long-term health problems.

These vaccines are used to protect against:

- Hib (*Haemophilus influenzae* type b) disease
- Hepatitis B
- HPV (Human papillomavirus)
- Whooping cough (part of the DTaP combined vaccine)
- Pneumococcal disease
- Meningococcal disease
- Shingles

4. Toxoid vaccines

Distinctive toxins produced by certain infective microbial organisms are utilized again to produce toxoid vaccines. This vaccine generates immunity tailored to the specific toxins rather than the causative micro-organism. Like some other types of vaccines, booster shots are required to get ongoing protection against diseases.

Toxoid vaccines are used to protect against:

- Diphtheria
- Tetanus

5. Viral vector vaccines

Viral vectors are artificially designed vehicles utilized to carry a foreign genetic material into another cell. It uses a modified version of a vector to deliver immunity. Scientists have long been studying viral vector vaccines of which in the recent past outbreak of Ebola has utilized the viral vector technology. A good number of studies focusing on other infectious diseases such as Zika, flu, and HIV infections are done, but yet to develop viral vector vaccines. Several different viruses have been used as vectors, including influenza, vesicular stomatitis virus (VSV), measles virus, and adenovirus, which causes the common cold.

Adenovirus is one of the viral vectors used to insert the spike protein of COVID-19 to produce vaccine (Oxford/Astra-Zeneca) that is now in phase 3 of the clinical trials.

6. DNA and RNA vaccines

DNA and RNA vaccines have been a subject of research for decade. A recent approach to immunization has used DNA containing a sequence of the pathogen's genome. This DNA is typically a bacterial plasmid engineered to include the sequence of an antigenic protein from the pathogen. This DNA can enter a number of cell types, and it can be expressed there using cellular transcription and translation machinery. In this respect, DNA vaccines act much like viruses. However, these DNAs contain only a very limited amount of genetic information and cannot become infectious. The mechanisms of uptake and induction of the immune response are not yet clear.

Nonetheless, promising results have been observed against viruses, bacteria, and parasitic microorganisms. This approach holds great promise for the development of effective vaccines against intractable diseases including HIV/AIDS, tuberculosis, and malaria.

As such with the Messenger Ribonucleic Acid (mRNA) vaccine that are recently being developed and trialed for the coronavirus (SARS-CoV-2) pandemic. These vaccines are genetically engineered to make proteins resembling that of the microbe as in the spike protein of SARS-CoV-2. The benefits of these vaccines include a shorter manufacturing time and no risk at all of causing disease as in live vaccines. However, there are risks with genetic manipulation or mutation in the host as the vaccine enters into the host cell in order to use its machinery for production of immunity.

COVID-19 VACCINES

Immune Response to COVID-19 and Vaccine Development

The immune response to the SARS-CoV-2 involves innate immune activation and antigen-specific responses of B and T cells as describe previously. Protection from viral infection is mainly achieved by virus-neutralizing antibodies, a principle that applies to the vast majority of viral infections to which humans acquire robust immune protection due to infection or vaccination. It is urgent to develop vaccines aiming at the induction of protective immune responses, primarily through virus-neutralizing antibodies specific for SARS-CoV-2.

Although at least 1–2 years are required to make effective vaccines available globally, vaccination may still be the most rapid and economical strategy to achieve widespread immune protection (herd immunity). Several organizations, including WHO, the Coalition for Epidemic Preparedness Innovations (CEPI) and the Global Vaccine Alliance International (GAVI), are increasing preparation efforts for future pandemics because of the lessons learned from SARS-CoV-2. Nevertheless, the challenge in developing vaccines still remains the same as microbes are still evolving and as technology continues to advance.

What are the current COVID-19 vaccines available? And describe their status, use, recommended group for administration, dosage, common side effects and how they work?

1. Pfizer-BioNTech

In the United States of America (USA), Pfizer-BioNTech vaccine was the first COVID-19 vaccine to receive a Food and Drug Administration (FDA) Emergency Use Authorization (EUA) after positive clinical trial data on December 11, 2020.

On December 11, 2020, this became the first COVID-19 vaccine to receive a Food and Drug Administration (FDA) Emergency Use Authorization (EUA), after the company reported positive clinical trial data, which included news that the vaccine was up to 95% effective at preventing symptomatic disease. But the Pfizer-BioNTech vaccine has had strict requirements involving how the vaccine is stored. For instance, it has required shipping in ultra-cold temperature-controlled units (-94 degrees Fahrenheit). In mid-February, the company submitted new data to the FDA demonstrating the stability of the vaccine at temperatures more commonly found in pharmaceutical refrigerators and freezers. Approval would make the vaccine easier to distribute.

Status: Emergency use in the U.S and also authorized for use in the European Union (under the name Comirnaty).

Recommended for: Anyone 16 and older. Adolescents and younger group are still yet to be tested.

Dosage: Two shots, 21 days apart

Common side effects: Chills, headache, pain, tiredness, and/or redness and swelling at the injection site, all of which generally resolve within a day or two of rest, hydration, and medications like acetaminophen. (If symptoms don't resolve within 72 hours or if you have respiratory symptoms, such as cough or shortness of breath, call your doctor.) On rare occasions (as in, 11 cases in 18 million vaccinations), mRNA vaccines have appeared to trigger anaphylaxis, a severe reaction that is treatable with epinephrine (the drug in Epipens®). For that reason, the Centers for Disease Control and Prevention (CDC) requires vaccination sites to monitor everyone for 15 minutes after their COVID-19 shot, and for 30 minutes if they have a history of severe allergies or are taking a blood thinner.

2. Moderna

Moderna's vaccine was the second one authorized for emergency use in the U.S. It received FDA EUA on December 18, 2020, about a week after the Pfizer vaccine. Moderna is also an mRNA vaccine, using the same technology as the Pfizer-BioNTech one and with a similarly high efficacy at preventing symptomatic disease.

There are two key differences:

- The Moderna vaccine can be shipped and kept in long-term storage in standard freezer temperatures, and stored for up to 30 days using normal refrigeration, making it easier to distribute and store.
- The Moderna vaccine was slightly less effective in clinical trials, about 86% in people who are 65 and older.

Status: Emergency use in the U.S and also authorized for use in the European Union.

Recommended for: Adults 18 and older. Moderna is still testing the vaccine in children ages 12-17, and children aged 6 months to 12 years.

Dosage: Two shots, 28 days apart

Common side effects: Similar to the Pfizer vaccine, side effects can include chills, headache, pain, tiredness, and/or redness and swelling at the injection site, all of which generally resolve within a day or two. On rare occasions, mRNA vaccines have appeared to trigger anaphylaxis, a severe reaction that is treatable with epinephrine (the drug in Epipens®). For that reason, the CDC requires vaccination sites to monitor everyone for 15 minutes after their COVID-19 shot and for 30 minutes if they have a history of severe allergies.

3. Johnson & Johnson/Janssen

On February 27, 2021, the FDA granted emergency use approval for a different type of vaccine, called a carrier, or virus vector vaccine. In early April, the CDC and FDA issued a joint recommendation for states to halt use of the Johnson & Johnson vaccine “out of an abundance of caution” during an investigation into reports of six rare, but serious clotting problems among women ages 18 to 48, occurring six to 13 days after vaccination (the recommendation fell short of an order to stop using the vaccine, leaving final decisions to the individual states).

But on Friday, April 23, the Food and Drug Administration (FDA) ended its recommended pause on the vaccine and will add a warning label about an uncommon, but potentially serious, blood clotting disorder. This decision followed a vote by a panel of advisers to the Centers for Disease Control and Prevention (CDC) to end the pause. This clears the way for states to resume vaccinations with the Johnson & Johnson vaccine.

In comparison to the Pfizer and Moderna vaccines, this one is easier to store (in refrigerator temperature), and requires only a single shot, which has made it easier to distribute and administer. An analysis released by the FDA in late February showed that the vaccine may reduce the spread of the virus by vaccinated people.

Status: Emergency use in the U.S.; authorized for use in the European Union (under the name Janssen).

Recommended for: Adults 18 and older. The company also expects to start testing the vaccine on children.

Dosage: Single shot. In November, Johnson & Johnson announced it would launch a second Phase 3 clinical trial to study using two doses, two months apart, to see if that regimen will provide better protection.

Common side effects: Fatigue, fever headache, injection site pain, or myalgia (pain in a muscle or group of muscles), all of which generally resolve within a day or two. It has had noticeably milder side effects than the Pfizer and Moderna vaccines, according to the FDA report released in late February. No one suffered an allergic reaction in clinical trials for the vaccine, according to the company.

4. Oxford-AstraZeneca

This vaccine, which is currently being distributed in the United Kingdom and other countries, is distinguished from some of its competitors by its lower cost (it's cheaper to make per dose), and it can be stored, transported, and handled in normal refrigeration for at least six months. Some countries temporarily suspended use of this vaccine in March after a small number of recipients developed blood clots. In April, a European Medicines Agency (EMA) safety committee concluded “unusual blood clots with low blood platelets should be listed as very rare side

effects" that could occur within two weeks of receiving the vaccine. While the U.K. called for further investigation, EMA regulators stressed that the benefits of the vaccine still outweigh the risks.

Status: Not available in the U.S., authorized for use in the European Union (under the name Vaxzevria).

Recommended for: Adults 18 and older

Dosage: Two doses, four to 12 weeks apart

Common side effects: Tenderness, pain, warmth, redness, itching, swelling or bruising at the injection site, all of which generally resolve within a day or two.

5. Chinese Vaccine: Sinopharm and Sinovac

On May 7, 2021, the Chinese vaccine was approved by the WHO for use in the 3rd World countries. It is the first vaccine developed by a non-Western country to get WHO backing. After considering its safety, efficacy aquality, the WHO has approved of its use because of the potential to rapidly accelerate Covid-19 vaccine access for countries seeking to protect health workers and populations at risk. The two Chinese vaccines (Sinopharm and Sinovac) differ significantly from some of the other Covid vaccines currently in use, especially those by Pfizer and Moderna. They are inactivated vaccines which were developed in a more traditional way where vaccines were being developed in the past. The killed viral particles are used to expose to the immune system to the virus without risking a serious disease response.

Status: Recently authorized by WHO for emergency use in low and middle income countries who cannot afford for other vaccines.

Recommended for: 18 years and older

Dosage: two doses

Common Side Effects: Noticed locally around the injection site there are flashes, itching and erythema (reddening and inflammation) formation like other vaccines. Systemically common adverse effects include fever, fatigue, muscle ache, joint pain, cough, difficulty breathing, nausea, diarrhea, and itchy skin. Other uncommon side effects include dizziness, loss of appetite, vomiting, oropharyngeal pain, difficulty swallowing, runny nose, constipation, and hypersensitivity. For rare, it lists acute allergic reaction, lethargy, drowsiness, difficulty sleeping, sneezing, nasopharyngitis, nasal congestion, dry throat, influenza, hypoesthesia, limb pain, palpitations, stomach ache, rash, skin and mucous membrane abnormalities, acne, eye pain, ear discomfort, and lymphadenopathy.

Table HM-IV1: Summary of Approved Vaccines for COVID-19 and how they work.

Approved Vaccines In Use	How the Vaccines work
1. Pfizer-BioNTech	This is a messenger RNA (mRNA) vaccine. The Pfizer-BioNTech mRNA vaccine delivers a tiny piece of genetic code from the SARS CoV-2 virus to host cells in the body, essentially giving those cells instructions, or blueprints, for making copies of spike proteins. The spikes do the work of penetrating and infecting host cells. These

	proteins stimulate an immune response, producing antibodies and developing memory cells that will recognize and respond if the body is infected with the actual virus.
2. Moderna	Similar to the Pfizer vaccine, this is an mRNA vaccine that sends the body's cells instructions for making a spike protein that will train the immune system to recognize it. The immune system will then attack the spike protein the next time it sees one (attached to a real SARS CoV-2 virus).
3. Johnson & Johnson	This is a viral vector vaccine also known as a carrier vaccine. Scientists engineer a harmless adenovirus (a common virus that, when not inactivated, can cause colds, bronchitis, and other illnesses) as a shell to carry genetic code on the spike proteins to the cells (similar to a Trojan Horse). The shell and the code are non-infective, but once the code is inside the cells, the cells produce a spike protein to prompt the body's immune system to create antibodies and memory cells to protect against an actual SARS-CoV-2 infection.
4. Oxford/Astra-Zeneca	Similar to the Johnson & Johnson's vaccine, this is a viral vector or carrier vaccine, made from a modified version of a harmless adenovirus. The final product contains the spike protein found in SARS-CoV-2. When that protein reaches the body's cells, the immune system mounts a defense, creating antibodies and memory cells to protect against an actual SARS-Cov2 infection.
5. Chinese: Sinopharm and Sinovac	This are inactivated (killed) vaccines where the killed viral particles (spike protein) is being exposed to the adaptive immune system for the B cells to produce antibodies against the virus. In the process, T cells create memory cells for the long term immunity against COVID-19 in the case of re-infection.

Note: Vaccines are continuously being developed and some are still under clinical trials. Imminently the **Chinese Sinovac, Russia's Sputnik, US Novavax** vaccines are still under assessment and validation by the WHO for use.

REFERENCES:

- Difference between Vaccination and Immunization. BYJU'S.
(Online: <https://byjus.com/biology/difference-between-vaccination-and-immunization/>)
- Schulz S M. Proteins 2: Structure-function relationship in protein families. Devlin M T- Textbook of biochemistry with clinical correlations. John Wiley and Sons. 2010.
- Vaccine types. Office of Infectious Disease and HIV/AIDS Policy, Department of Health & Human Services. April 29, 2021. U.S.
(Online: <https://www.hhs.gov/immunization/basics/types/index.html>)
- Speiser D E, Beckmann M F. COVID-19: Mechanisms of Vaccination and Immunity. *Vaccines* **2020**, 8(3), 404; (Online: <https://doi.org/10.3390/vaccines8030404>)
- Siegrist C. Vaccine Immunology.
(Online: https://www.who.int/immunization/documents/Elsevier_Vaccine_immunology.)
- Understanding How Vaccines Work. (Online: www.cdc.gov/vaccines/conversations)

- Katella K. Comparing the COVID-19 Vaccines: How Are They Different? Family health, Yale Medicine. MAY 3, 2021. (Online: <https://www.yalemedicine.org/news/covid-19-vaccine-comparison>)
- BBC world news. Sinopharm: Chinese Covid vaccine gets WHO emergency approval. 8 May 2020. (<https://www.bbc.com/news/world-asia-china-56967973>)
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Topic 3: Public Health

Activity 1: Case Study

Title: COVID-19 (SARS-CoV2-2019) Pandemic

Background

The last decade has witnessed the outbreak of many life-threatening human pathogens including *Nipah*, *Ebola*, *Chikungunya*, *Zika*, *Middle East respiratory syndrome coronavirus (MERS-CoV)*, *Severe Acute respiratory syndrome coronavirus (SARS-CoV)* and more recently **novel coronavirus (2019-nCoV or SARS-CoV-2)**. The disease condition associated with novel coronavirus is referred to as Coronavirus disease (COVID-19). The emergence of novel coronavirus in 2019 in Wuhan, China marked the third highly pathogenic coronavirus infecting humans in the 21st century. The continuing emergence of coronaviruses at regular intervals poses a significant threat to human health and economy. Not only does it affect the health and economy, but it also affects the social and psychological well-being of individuals as well as reflecting the governance and leadership provided by different countries.

The COVID-19 pandemic holds great challenges for which the world is only partially prepared. The SARS-CoV-2 infection is very contagious and infectious with spread from both symptomatic patients as well as pre-symptomatic or asymptomatic carriers. To limit the spread of COVID-19, aggressive measures of enforcing and preventing spread of infection such as pandemic acts with lock downs and primary efforts focusing on isolation or confinement, with physical or social distancing (1 – 2 metres), hygiene measures such as hand wash techniques with alcohol or soap and face masks (N95 respirators).

At present, many investigations aim at defining optimal strategies to limit viral transmission while simultaneously permitting business and social activities to continue. Scientific insights and understanding of the biological mechanisms of the virus and its capability to spread are still primordial. Built on this knowledge, the practical strategies may have at least set three priorities:

- Firstly, to continue hygiene measures and physical or social distancing;
- Secondly, to maximize the viral monitoring, both geographically and in time to focus viral containment locally and limit transmission wherever and whenever possible;
- Thirdly, to rapidly increase the immunity of the world's population.

The SARS-CoV-2 virus prefers the environment surrounding the oral cavity and nasopharyngeal region due to the moisture and mucous membranes. Hence, nasopharyngeal or salivary swabs are taken for the diagnosis of SARS-CoV-2 infection through detection of viral RNA by nucleic acid tests (NATs) or tests that detect viral protein antigens. In infected individuals, the results are only positive for a relatively short time window, on average until the 14th day after symptom onset. Furthermore, a positive NAT result does not allow scientists to conclude whether the affected person is or will become immune. Therefore, serological tests are needed, as they can detect the various types of antibodies in the blood that persist for months or even years.

Health Impacts

The health implications of COVID-19 are now very clear. It has tested the capability, effectiveness and efficiency of different health systems throughout the world. The varying health

systems are gearing up to filling in the loop holes within their health systems as well as researching into providing new therapeutic/prophylactic interventions and/or vaccines to reduce the spread of COVID-19. This pandemic has prompted tenacious studies at different institutions with multiple publications throughout world to investigate and understand the causative organism, its structure, function, characteristics, and pathogenesis in order to develop therapeutic interventions to reduce or stop its spread.

Ironically, even after a decade of research on coronavirus, there has not been any licensed vaccine or therapeutic agents to treat coronavirus infection. Till today, new vaccines are being rapidly developed and trialed for COVID-19 within this short span of time after the outbreak in 2019. It is the first time in the global history that an mRNA (messenger Ribonucleic Acid) vaccine is being developed for a virus. There have been several infectious diseases that were around for some time with years of research, but no vaccines have been developed to fight them such as HIV/AIDS, malaria and tuberculosis (TB). Several inquiries into how the vaccines work, their efficacy, adverse effects and what population groups are at risk and should be vaccinated are now the challenge faced by the scientific community in developing new therapeutic interventions.

Papua New Guinea is a classical case where there is a need to develop a mechanistic understanding of immunity against SARS-CoV-2. It will be important to determine to which degree natural infection induces immunity and for how long it can protect from re-infection. It may take significantly longer than one year until the majority of the population acquire immunity through infection. More focus should be on the development and validation of precision serology and research on immune responses induced by natural infection. The basic and applied knowledge acquired during the last few decades on coronaviruses is highly useful to perform serological studies on why Papua New Guineans are not dying like flies, but are showing increased likelihood of inducing natural immunity against SARS-CoV-2 infection. Therefore in avoiding adverse effects and harmful disease enhancement from the vaccines recently being developed and trialed, it is now, probably more than ever, the world depends on rational effective strategies built on robust scientific evidence.

In addition, since the COVID-19 pandemic, much attention has been focused on it while other impending fatal communicable diseases have been given less attention. The Global Fund to Fight AIDS, Tuberculosis and Malaria showed COVID-19 has massively disrupted health systems and health service delivery for HIV, TB and malaria in low- and middle-income countries in Africa and Asia in 2020. Not only this communicable (infectious) disease, but also other communicable diseases and the non-communicable or lifestyle diseases also contribute to the increasing mortality (death rate) globally. Therefore, it is recommended for the Health Department to urgently scale up the adaptive measures that health facilities adopted to continue the fight against all these other killer diseases to ramp up delivery of critical supplies for the COVID-19 response, and prevent health care systems and community responses from collapsing.

Economic Impacts

COVID-19 is not only a global pandemic and public health crisis; it has also severely affected the global economy and financial markets. Significant reductions in income, a rise in unemployment, and disruptions in the transportation, service, and manufacturing industries are among the consequences of the disease mitigation measures that have been implemented in many countries. It has become clear that most governments in the world underestimated the risks of rapid COVID-19 spread and were mostly reactive in their crisis response. As disease

outbreaks are not likely to disappear in the near future, proactive international actions are required to not only save lives but also protect economic prosperity.

The COVID-19 pandemic has caused a break in the global supply chain affecting all economies throughout the world. And PNG is not isolated with a tiny virus shaking the world; PNG is ill prepared in terms of technology and facilities to cater for such pandemics. It is now a wakeup call to our government not only to solve such health issues, but also to invest in and promote research, science and technology within the country.

Below is an informative and well-presented article of current COVID-19 Pandemic in Papua New Guinea regarding mostly the economic impacts and livelihood of the people written by **Dr. Elizabeth Kopel** (Senior Research Fellow and Program Leader of the Informal Economy Research Program at the Papua New Guinea National Research Institute) titled:

‘The impact of COVID-19 on livelihoods in Papua New Guinea’

“The swift adoption of a State of Emergency (SoE) in response to COVID-19 was a crucial measure to prevent the spread of the virus in PNG. However, restrictions imposed by the SoE have had a devastating impact on the majority of people whose livelihoods are strongly linked to informal economic activities.

Restrictions imposed on travel and closure of businesses during the SoE limited, and in some cases prevented, access to essential goods and services with major impacts on food supply chains. While some shops and markets were permitted to remain open, police have not adequately facilitated implementation, impacting the livelihoods of both rural and urban households. Rural produce has not reached urban markets, similarly urban supplies have not reached rural and outlying provinces. Limited supply drives prices up, making crucial supplies too expensive for ordinary citizens.

Restrictions have also led to job losses and a drastic fall in incomes. This has disproportionately affected a higher proportion of households that are already disadvantaged, both economically and socially. The poor cannot stock up on food and have no savings for emergencies so without work there is no money. This translates to thousands of informal enterprises and livelihoods lost.

Less disposable income makes it harder to afford basic needs for survival. Most people struggle to meet daily needs in ordinary times, so it is extremely hard for households to cope for weeks and or even months without a regular income. Many are going without food and this is a catalyst for creating additional social issues: domestic violence over household finances; increased theft; and petty crime driven by hunger and starvation. Poor diet and nutrition weaken the body’s natural defense system and leave the poor and disadvantaged susceptible to disease.

Fortunately, the country has been lucky to escape with under 10 confirmed cases of COVID-19 and no deaths up to now, which has led to the relaxation of some restrictions. Current social unrest in Peru and Brazil over loss of jobs, income, hunger, and starvation due to lockdowns highlight the potential risks ongoing restrictions pose for livelihoods.

Urban settlements are home to low-income workers and the majority of those are dependent on the informal economy for livelihoods. Most residents live in poor and overcrowded housing conditions without adequate access to clean water, sanitation, and hygiene. These factors would make it easy for COVID-19 to spread in these communities.

Heavy-handed policing in the administration of the SoE without flexibility for the movement of essential goods, services, and for people to access basic services has led to ongoing public complaints. Destruction and confiscation of goods, as well as collection fees for access through checkpoints, were commonly reported issues, yet such actions are contrary to the purpose and function of policing.

All this has greatly affected the lives of individuals and families not only in PNG, but also the third world countries throughout the world. It is now crucial to be economically prepared to sustain such crisis. ¹⁴

Social and Psychological Impacts

The COVID-19 pandemic and the resulting economic recession have negatively affected many people's mental health and created new barriers for people already suffering from mental illness and substance use disorders. As the pandemic wears on, ongoing and necessary public health measures expose many people to experiencing situations linked to poor mental health outcomes, such as isolation and job loss.

The current COVID-19 pandemic has instilled and up roared fear amongst citizens via the mainstream media as well as social networking sites such as facebook, twitter, etc. People are now afraid of dying, unlike before people used to live normally without fearing death. This new normal is a psychological reset from our modus operandi and is now taking shape with force. Instead of advocating for a healthy living and prevention of COVID-19 which is here to stay, the health experts are continuously debating and grumbling amongst themselves over issues beyond our technology and control.

The pandemic has both short- and long-term implications for mental health and substance use, particularly for groups at risk of new or exacerbated mental health disorders and those facing barriers to accessing care. Phased COVID-19 vaccinations are taking place across the country, perhaps signaling that the end of the pandemic is on the horizon. However, many of the stressful conditions employed to mitigate the spread of the coronavirus are likely to persist for the near future, given the slow and troubled rollout of vaccinations across the country, instances of people refusing the vaccine due to fear or uncertainty, and the need for vaccinated people to continue taking existing precautions to mitigate the outbreak.

History has shown that the mental health impact of disasters outlasts the physical impact, suggesting today's elevated mental health need will continue well beyond the coronavirus outbreak itself. For example, an analysis of the psychological toll on health care providers during outbreaks found that psychological distress can last up to three years after an outbreak. Due to the financial crisis accompanying the pandemic, there are also significant implications for mortality due to "deaths of despair." A May 2020 analysis projects that, based on the economic downturn and social isolation, additional deaths due to suicide and alcohol or drug misuse may occur by 2029.

As policymakers continue to discuss further actions to alleviate the burdens of the COVID-19 pandemic, it will be important to consider how the increased need for mental health and

¹⁴Elizabeth Kopel. The impact of COVID-19 on livelihoods in Papua New Guinea: Can PNG fight the coronavirus crisis whilst protecting its most vulnerable citizens? Asia and pacific policy forum. (<https://www.policyforum.net/the-impact-of-covid-19-on-livelihoods-in-papua-new-guinea/>)

substance use services will likely persist long term, even if new cases and deaths due to the novel coronavirus subside.¹⁵

Political Decision Making

Politics plays a key role since governance came into existence beginning from Heaven down to Earth. It is about governance and good leadership that will guide a nation in such terror and suffering the world is suffering today. It is now up to different stakeholders of a country to step up and take stage in making informed decisions for the benefit of their people in the course of such crisis. It is during crisis that raises a True Wise Leader who finds opportunity in making decisions that changes the tide of such health and economic crisis.

So far, decisions made by the government might not be as fulfilling and promising for the nation of PNG. However, leaders are chosen as shepherd's to lead and guide the sheep's. Therefore, whether we are led to the slaughter house or across to green pastures relies heavily on the political decision making by our leaders.

Recommendations for Policy Makers (Continuation of Article by Dr. Elizabeth Kopel)

"In response to these issues, there are a number of strategies PNG can implement to reduce negative impacts of COVID-19 and safeguard livelihoods.

The first strategy should be to ensure that markets and shops are open for access to essential goods and services. Informal markets provide the lifeline for a majority of people; income for market vendors and access to locally grown food for consumers. It is crucial to ensure that fresh food from rural and peri-urban producers is collected and safely enters urban markets.

Similarly, it is vital for shops to remain open for people to access food and other essential supplies. It is even more important for goods to be transported to rural areas without the need for people to travel into towns. This requires giving the private sector easy access to loans for establishing businesses providing transportation and storage services or local commercial infrastructure in the outer districts. Services need to be available where people are instead of making them travel long distances.

The second strategy should be to ensure sensible policing. Policing must be done with respect and understanding to facilitate easy movement of people, food, and other essential goods and services, of course with observation of social distancing and hygiene rules. A ruthless style of policing creates division, resentment, and conflict.

Third, the PNG government needs to facilitate the provision of alternative income streams for groups at high risk. For example, employment of betelnut vendors to clean and beautify suburbs or areas where they would normally be selling, so that they can earn an income as well as keeping the towns and cities clean. Incentives can also be created for betelnut growers to start planting other crops which would provide a steady flow of income.

¹⁵ Nirmita Panchal , Rabah Kamal , Cynthia Cox, and Rache Garfield. The Implications of COVID-19 for Mental Health and Substance Use. Feb 10, 2021 (The Implications of COVID-19 for Mental Health and Substance Use | KFF) [www.kff.org](https://www.kff.org/coronavirus-covid-19/issue-brief/the-implications-of-covid-19-for-mental-health-and-substance-use/) [https://www.kff.org/coronavirus-covid-19/issue-brief/the-implications-of-covid-19-for-mental-health-and-substance-use/]

Fourth, municipal authorities and market management should be responsible for markets. The supervision and implementation of hygiene and social distancing measures to prevent the spread of COVID-19 in markets can be done by municipal authorities, market managers, and vendor representatives. SoE management needs to ensure that trained public health officers provide appropriate training of supervisors.

As a fifth strategy, the government should utilise existing community networks for awareness-raising in settlements. Awareness activities to reach out to urban settlement communities is vital to mass educate the bulk of the population on prevention of COVID-19 when conducting their economic activities as well as in the home. Awareness needs to be conducted through existing networks and organisations. This will create ownership, retain knowledge within respective communities, and improve sustainability.

The sixth and final strategy should be to enable churches to play a bigger role in addressing COVID-19 to reach rural communities. Churches play an important role in the country as they operate in places where even state agencies are not actively present. Allocation of resources to churches for conducting COVID-19 awareness activities in rural areas would achieve much in terms of reaching the length and breadth of the country at a cheaper cost.

It is important that policymakers enable appropriate intervention strategies to reduce these challenges and ensure that livelihoods are safeguarded whilst keeping the virus at bay.”¹⁶

Application of Knowledge, Skills and Values (AKSA)

1. The knowledge gained in health and medicine would focus students on topical health issues such as COVID-19 pandemic and to prepare for such in the near future.
2. Students would be sufficiently versed and equipped to challenge or query medical opinions and general health decision making that would impact their lives.

¹⁶ Elizabeth Kopel. The impact of COVID-19 on livelihoods in Papua New Guinea: Can PNG fight the coronavirus crisis whilst protecting its most vulnerable citizens? Asia and Pacific Policy Forum. (<https://www.policyforum.net/the-impact-of-covid-19-on-livelihoods-in-papua-new-guinea/>)

REFERENCES:

Biodiversity

- Bolger J. Sciencing: four types of biodiversity. Leaf Group Ltd. March 09, 2018. (Accessed online 18-02-21: <https://sciencing.com/difference-between-community-ecosystem-5776576.html>)
- Convention on biological diversity. What is Agricultural Biodiversity? UN environment programme. (Accessed online 15-02-21: <https://www.cbd.int/agro>)
- Biodiversity: What is it, where is it, and why is it important? (www.greenfacts.org)
- Four Types of Biodiversity (sciencing.com)
- 3 Types of Biodiversity - World of Biodiversity (worldofbiodiversity.weebly.com)
- COP Decision (www.cbd.int. Feb 9, 2021)
- What is Agricultural Biodiversity? (www.cbd.int. Feb 9, 2021)
- Megadiverse countries - Wikipedia (en.wikipedia.org)
- megadiverse-countries.pdf (www.biodiversitya-z.org)
- Conservation of Megadiverse Countries - Iberdrola (www.iberdrola.com)
- Biodiversity in Papua New Guinea (www.coolearth.org)
- https://www.bascompte.net/content/publications/Camara_Leret_Nature_20.pdf (www.bascompte.net)
- Native trees of Mexico: diversity, distribution, uses and conservation [PeerJ] (peerj.com)
- New Guinea has the most plant species of any island (news.mongabay.com)
- World Programme for the Census of Agriculture - Wikipedia (en.wikipedia.org)
- Home | Food and Agriculture Organization of the United Nations (www.fao.org)
- The link between ant agriculture and early social complexity | Seshat: Global History Databank (seshatdatabank.info)
- Species under threat from exploitation in New Guinea | WWF (wwf.panda.org)
- Papua New Guinea Threatened plant species, 1960-2020 - knoema.com (knoema.com)
- Wildlife of Papua New Guinea - Woodland Park Zoo Seattle WA (www.zoo.o)
- E2 Flora and Fauna of PNG Student.doc (www.education.gov.pg)
- Status of management of plant and animal genetic resources in Papua New Guinea | Papua New Guinea Data Portal (png-data.sprep.org)
- Plant genetic resources of PNG_ Some thoughts on intellectual property rights.pdf (png-data.sprep.org)
- Making the most of Papua New Guinea's biodiversity: Establishment of an integrated set of programs that link botanical survey with pharmacological assessment in "The Land of the Unexpected" (www.ncbi.nlm.nih.gov)
- Papua New Guinea: A centre of genetic diversity in winged bean (*Psophocarpus tetragonolopus* (L.) D c.) | SpringerLink (link.springer.com)
- New Study Reveals Remarkable Genetic Diversity among Papuan New Guinean Peoples | Anthropology, Genetics | Sci-News.com (www.sci-news.com)
- Papua New Guinea: Productive Partnerships in Agriculture Project (www.worldbank.org)
- Food and Agriculture in Papua New Guinea (press.anu.edu.au)
- Papua New Guinean Agriculture: Significant Opportunity and Much Peril - Future Directions International (www.futuredirections.org.au)
- Agriculture - Papua New Guinea - export, crops, sector (www.nationsencyclopedia.com)
- Agriculture in Papua New Guinea - Wikipedia (en.wikipedia.org)

- [New Agriculturist: Country profile - Papua New Guinea](http://www.new-ag.info) (www.new-ag.info)
- [The Emergence of Scientific Reasoning | IntechOpen](http://www.intechopen.com) (www.intechopen.com;www.sanger.ac.uk)
- [Papua New Guinea: Productive Partnerships in Agriculture Project](http://www.worldbank.org) (www.worldbank.org)
- [Papua New Guinean Agriculture: Significant Opportunity and Much Peril - Future Directions International](http://www.futuredirections.org.au) (www.futuredirections.org.au)
- [Agriculture - Papua New Guinea - export, crops, sector](http://www.nationsencyclopedia.com) (www.nationsencyclopedia.com)
- [Agriculture in Papua New Guinea - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- www.fao.org (www.fao.org)
- [New Agriculturist: Country profile - Papua New Guinea](http://www.new-ag.info) (www.new-ag.info)
- [Chapter 6.1: What is genetic diversity? - Animal breeding - Groen Kennisnet wiki](http://wiki.groenkennisnet.nl) (wiki.groenkennisnet.nl)
- [Assessment of genetic diversity in crop plants - an overview - MedCrave online](http://medcraveonline.com) (medcraveonline.com)
- [Importance of Genetic Diversity Assessment in Crop Plants and Its Recent Advances: An Overview of Its Analytical Perspectives](http://www.hindawi.com) (www.hindawi.com)
- [Genetic Diversity in Plants | IntechOpen](http://www.intechopen.com) (www.intechopen.com)
- [Agriculture classification of crops - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- [taxonomy of agricultural crops - Google Scholar](http://scholar.google.com) (scholar.google.com)
- [Agricultural biodiversity - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- [What is Agrobiodiversity?](http://www.fao.org) (www.fao.org)
- [Fe'i banana - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- [Musa troglodytarum - Useful Tropical Plants](http://tropical.theferns.info) (tropical.theferns.info)
- [Fe'i Bananas Information, Recipes and Facts](http://specialtyproduce.com) (specialtyproduce.com)
- [Fei bananas | Improving the understanding of banana](http://www.promusa.org) (www.promusa.org)
- [Phenotypic variation of Fei Banana \(Musa Troglodytarum L.\) originated from Maluku Islands | Request PDF](http://www.researchgate.net) (www.researchgate.net)
- [Golden bananas in the field: elevated fruit pro-vitamin A from the expression of a single banana transgene - Paul - 2017 - Plant Biotechnology Journal - Wiley Online Library](http://onlinelibrary.wiley.com) (onlinelibrary.wiley.com)
- [File:Fe'i banana map.png - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- [Annual Report 2017 Bioversity International](http://www.biodiversityinternational.org) (www.biodiversityinternational.org)
- [Golden bananas high in pro-vitamin A developed: Research has produced a golden-orange fleshed banana, rich in pro-vitamin A -- ScienceDaily](http://www.sciencedaily.com) (www.sciencedaily.com)
- [carotene, iron and zinc content in Papua New Guinea and East African highland bananas.](http://www.researchgate.net) (www.researchgate.net)
- [PNG banana helps the fight against vitamin A deficiency around the world - The National](http://www.thenational.com.pg) (www.thenational.com.pg)
- [QUT - Institute for Future Environments - QUT develops golden bananas high in pro-vitamin A](http://www.qut.edu.au) (www.qut.edu.au)
- [Aus scientists share GM banana with India](http://www.fruitnet.com) (www.fruitnet.com)
- [Annual Report 2017 Bioversity International](http://www.biodiversityinternational.org) (www.biodiversityinternational.org)
- [The amazing diversity of bananas in Papua New Guinea](http://www.biodiversityinternational.org) (www.biodiversityinternational.org)
- [Vitamin-A rich GMO banana, which could fight malnutrition in Africa, ready for field trials | Genetic Literacy Project](http://geneticliteracyproject.org) (geneticliteracyproject.org)

- https://iris.wpro.who.int/bitstream/handle/10665.1/5398/9789290612490_eng.pdf (iris.wpro.who.int)
- [Health benefits of Red fruit \(Buah merah\) | HB times](http://www.healthbenefitstimes.com) (www.healthbenefitstimes.com)
- [Promoting \(pro\) vitamin A-rich bananas: A chronology - CGIAR](http://www.cgiar.org) (www.cgiar.org)
- [Research on micronutrient deficiencies using diet diversity](http://www.bioversityinternational.org) (www.bioversityinternational.org)
- [Birds of Paradise | National Geographic](http://www.nationalgeographic.com) (www.nationalgeographic.com)
- [Calla vs. Peace Lily](http://homeguides.sfgate.com) (homeguides.sfgate.com)
- [Bird-of-paradise | San Diego Zoo Animals & Plants](http://animals.sandiegozoo.org) (animals.sandiegozoo.org)
- [Bird-of-paradise - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- [Birds of Paradise](http://www.timlaman.com) (www.timlaman.com)
- [Birds of paradise Photography Art | Tim Laman](http://www.timlamanfineart.com) (www.timlamanfineart.com)
- [Brood parasite - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- [The Ecology of Brood Parasitism in Birds on JSTOR](http://www.jstor.org) (www.jstor.org)
- [Brood Parasitism](http://web.stanford.edu) (web.stanford.edu)
- [The coevolutionary biology of brood parasitism: a call for integration | Philosophical Transactions of the Royal Society B: Biological Sciences](http://royalsocietypublishing.org) (royalsocietypublishing.org)
- [Victoria Crowned-pigeon \(Goura victoria\) - BirdLife species factsheet](http://datazone.birdlife.org) (datazone.birdlife.org)

Biotechnology

- [Introduction to Biotechnology](http://www.researchgate.net) (www.researchgate.net)
- [Application of recombinant DNA technology \(genetically modified organisms\) to the advancement of agriculture, medicine, bioremediation and biotechnology industries - MedCrave online](http://medcraveonline.com) (medcraveonline.com)
- [FDA on transgenic animals—a dog's breakfast? | Learn Science at Scitable](http://www.nature.com) (www.nature.com)
- [Introduction of a rabbit \$\beta\$ -globin gene into the mouse germ line | Learn Science at Scitable](http://www.nature.com) (www.nature.com)
- [Recombinant DNA Technology and Transgenic Animals | Learn Science at Scitable](http://www.nature.com) (www.nature.com)
- [Molecular cloning - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- [Recombinant DNA - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- [recombinant DNA | Definition, Steps, Examples, & Invention | Britannica](http://www.britannica.com) (www.britannica.com)
- [Aggressive mimicry - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- [https://www.fda.gov/media/137450/download](http://www.fda.gov/media/137450/download) (www.fda.gov. Apr 5, 2021).
- [Agricultural Biotechnology | FDA](http://www.fda.gov) (www.fda.gov. Feb 8, 2021)
- [Types of Genetic Modification Methods for Crops | FDA](http://www.fda.gov) (www.fda.gov. Feb 8, 2021)
- [Biological Control of Plant Disease](http://www.researchgate.net) (www.researchgate.net)
- [Role of biotechnology in insect-pests management](http://www.researchgate.net) (www.researchgate.net)
- [Application of biotechnology](http://www.slideshare.net) (www.slideshare.net)
- [Pharmaceutical Biotechnology | Industrial Microbiology – 3rd International Conference on Microbiology and Infectious diseases](http://microbiologyplusconference.wordpress.com) (microbiologyplusconference.wordpress.com)
- [Industrial Biotechnology: An Overview](http://www.researchgate.net) (www.researchgate.net)
- [Industrial Biotechnology | Mary Ann Liebert, Inc., publishers](http://home.liebertpub.com) (home.liebertpub.com)
- [Medical Biotechnology – 3rd International Conference on Microbiology and Infectious diseases](http://microbiologyplusconference.wordpress.com) (microbiologyplusconference.wordpress.com)

- [Estimating the biotech sector's contribution to the US economy | Nature Biotechnology](http://www.nature.com) (www.nature.com)
- [Role of biotechnology in insect-pests management](http://www.researchgate.net) (www.researchgate.net)
- [Biological Control of Plant Disease](http://www.researchgate.net) (www.researchgate.net)
- [Special issue: Application of biotechnology for biofuels: transforming biomass to biofuels](http://www.ncbi.nlm.nih.gov) (www.ncbi.nlm.nih.gov)
- [Biotech and Pharmaceutical | WSP](http://www.wsp.com) (www.wsp.com)
- [Biotechnology - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- [How Medical Biotechnology Is Advancing Modern Healthcare](http://online.campbellsville.edu) (online.campbellsville.edu)
- [Medical Biotechnology: Techniques and Applications - ScienceDirect](http://www.sciencedirect.com) (www.sciencedirect.com)
- [Medical Biotechnology: Advancements And Ethics](http://www.wgu.edu) (www.wgu.edu)
- [Recombinant DNA - Isolating the clone | Britannica](http://www.britannica.com) (www.britannica.com)
- [recombinant DNA | Definition, Steps, Examples, & Invention | Britannica](http://www.britannica.com) (www.britannica.com)
- [Genetic Molecular Biology - ZapMeta](http://www.zapmeta.ws) (www.zapmeta.ws)
- [An Introduction to Recombinant DNA](http://www.rpi.edu) (www.rpi.edu)
- [Recombinant DNA Technology- Tools, Process, and Applications](http://byjus.com) (byjus.com)

Health and Medicine

- **References**
- Rang HP, Dale MM, Ritter JM, et al. Pharmacology (5th edition). London: Elsevier Science; 2003.
- Vadjia FJE. Generic substitution in epilepsy: A controversial issue [online]. The Epilepsy Report; 2006. [cited March 2009]. Available from: [\[URL link\]](#)
- Jones AW, Holmgren A, Kugelberg FC. Concentrations of scheduled prescription drugs in blood of impaired drivers: considerations for interpreting the results. *Therapeutic Drug Monitoring*. 2007; 29(2):248-60.
- Xie HG, Frueh FW. Pharmacogenomics steps toward personalised medicine. *Personalized Medicine*. 2005; 2(4): 325- 37.
- Bloche G. Race-based therapeutics. *N Engl J Med*. 2004; 351(20): 2035-7.
- Camí J, Farré M. Mechanisms of disease: Drug addiction. *N Engl J Med* 2003; 349:975-86.
- Briggs GG, Freeman RK, Yaffe S. Drugs in Pregnancy and Lactation: A Reference Guide to Fetal and Neonatal Risk (8th edition). Philadelphia: Lippincott Williams & Wilkins; 2008.
- Bedouch P, Allenet B, Grass A, et al. Drug-related problems in medical wards with a computerized physician order entry system. *Journal of Clinical Pharmacy and Therapeutics*. 2009; 34: 187-95.
- McBride AJ, Pates R, Ramadan R, et al. Delphi survey of experts' opinions on strategies used by community pharmacists to reduce over-the-counter drug misuse. *Addiction*. 2003; 98(4):487-97.
- Bessell TL, Anderson JN, Silagy CA, et al. Surfing, self-medicating and safety: buying non-prescription and complementary medicines via the internet. *Quality & Safety in Health Care*. 2003; 12(2):88-92.
- [Pharmacology - Wikipedia](http://en.wikipedia.org) (en.wikipedia.org)
- [Molecular Mechanisms of Drug Action: An Emerging View | Annual Review of Biophysics](http://www.annualreviews.org) (www.annualreviews.org)
- [Pharmacodynamics: Molecular Mechanisms of Drug Action | Goodman & Gilman's:](#)

- [The Pharmacological Basis of Therapeutics, 13e | AccessMedicine | McGraw-Hill Medical \(accessmedicine.mhmedical.co\)](#)
- [Pharmacodynamics: Molecular Mechanisms of Drug Action | Basicmedical Key \(basicmedicalkey.com\)](#)
- [Drug design and discovery - FIP - International Pharmaceutical Federation Drug design and discovery \(www.fip.org\)](#)
- [Drug design - Wikipedia \(en.wikipedia.org\)](#)
- [Pharmacodynamics: Molecular Mechanisms of Drug Action | Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 13e | AccessMedicine | McGraw-Hill Medical \(accessmedicine.mhmedical.com\)](#)
- [Molecular Mechanisms of Drug Action: An Emerging View | Annual Review of Biophysics \(www.annualreviews.org\)](#)
- [www.dsfarm.unipd.it www.dsfarm.unipd.it](#)
- [Pharmacodynamics - Wikipedia file:](#)
- [220px-Molecular_mechanisms.svg.png \(220×233\)](#)
- [The Radiology Assistant : Basic Interpretation \(radiologyassistant.nl\)](#)
- [Plain Film X-Ray - Principles - Interpretation – TeachMeAnatomy \(teachmeanatomy.info\)](#)
- [DIAGNOSIS OF A GENETIC DISEASE - Understanding Genetics - NCBI Bookshelf \(www.ncbi.nlm.nih.gov\)](#)
- [http://library.lol/main/F9974C06ADF10F1E8F63EF3E5CD92268library.lol](#)
- [Genetic Lab Testing - ZapMeta \(www.zapmeta.ws\)](#)
- [What are the types of genetic tests?: MedlinePlus Genetics \(medlineplus.gov\)](#)
- [DIAGNOSIS OF A GENETIC DISEASE - Understanding Genetics - NCBI Bookshelf \(www.ncbi.nlm.nih.gov\)](#)
- [NEWBORN SCREENING - Understanding Genetics - NCBI Bookshelf \(www.ncbi.nlm.nih.gov\)](#)
- [Interpretation of ultrasound - PubMedpubmed.\(ncbi.nlm.nih.gov\)](#)
- [Interpretation Of Ultrasound Image | KYAMC Journal \(www.banglajol.info\)](#)
- [www.radiologyinfo.orgwww.radiologyinfo.org](#)
- [GENETIC TESTING - Understanding Genetics - NCBI Bookshelf \(www.ncbi.nlm.nih.gov\)](#)
- [What is genetic testing?: MedlinePlus Genetics \(medlineplus.gov\)](#)
- [What Is A Genetic Testing - ZapMeta \(www.zapmeta.ws\)](#)
- [How is genetic testing done? MedlinePlus Genetics \(medlineplus.gov\)](#)
- [Genetic Testing | CDC \(www.cdc.gov\)](#)
- [https://www.bmj.com/content/bmj/370/bmj.m3325.full.pdf \(fwww.bmj.com\)](#)
- [DNA Genetic Testing & Analysis - 23andMe AU, DE, FR & EU \(www.23andme.com\)](#)
- [Shanmugaraj B, et al. Perspectives on monoclonal antibody therapy as potential therapeutic intervention for Coronavirus disease-19 \(COVID-19\). Asia Pacific J of Allergy and immunology. Asian Pac J Allergy Immunol 2020;38:10-18 \(DOI 10.12932/AP-200220-0773\)](#)
- [The Implications of COVID-19 for Mental Health and Substance Use | KFF \(www.kff.org\)](#)
- [The Indirect Health Effects of COVID-19: Emerging Findings from Kenya, the Philippines, South Africa, and Uganda | Center For Global Development \(www.cgdev.org\)](#)
- [Healthy young adults who had COVID-19 may have long-term impact on blood vessels and heart health -- ScienceDaily \(www.sciencedaily.com\)](#)

- The Impact Of The COVID-19 Pandemic On Hospital Admissions In The United States | Health Affairs (www.healthaffairs.org)
- The impact of COVID-19 on HIV, TB and malaria services and systems for health: A snapshot from 502 health facilities across Africa and Asia - World | ReliefWeb (reliefweb.int)
- Physical and mental health impacts of COVID-19 on healthcare workers: a scoping review (intjem.biomedcentral.com)
- Impact of COVID-19 on people's livelihoods, their health and our food systems (www.who.int.)
- The impact of COVID-19 on livelihoods in Papua New Guinea – Policy Forum (www.policyforum.net)
- PNG health system on verge of collapse following COVID-19 surge MSF (www.msf.org)